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An atom-photon quantum gate using an atomic clock qubit and a birefringent cavity

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Abstract

Single atoms coupled to optical cavities provide a powerful platform for quantum information processing. Minimizing the influence of external factors, which would otherwise compromise coherence and gate fidelity, is a key goal in this endeavor. For instance, spin qubits encoded in magnetically sensitive states suffer from dephasing due to ambient magnetic field fluctuations, motivating the use of magnetic-field-insensitive clock states. In this work, I investigate an atom-photon gate protocol that employs a single ^{87}Rb atom prepared in the clock states of the $5S_{1/2}$ hyperfine ground manifold coupled to a high-finesse birefringent Fabry–Pérot cavity. The cavity's high birefringence creates two well-separated polarization eigenmodes, of which only one is resonantly coupled to the atomic qubit transition. The reflectivity from the atom-cavity system thus becomes dependent on the state of the atom and the polarization of the incoming photon, with the atom acting as the control qubit and the photon as the target qubit. Using this, I implemented a controlled-phase-flip (CPF) gate between the atomic and the photonic qubits. By then preparing the photonic qubit in a superposition of the cavity's polarization eigenmodes, I realize an atom-photon controlled–NOT (CNOT) gate. I present simulations based on input–output theory and master-equation modeling that quantify the conditional reflection amplitudes and gate truth tables under realistic experimental imperfections such as finite mode matching, intracavity losses, multiphoton contributions, and atomic scattering. Experimentally, I measure an overlap of $(92.41 \pm 0.52)\,\%$ and $(88.15 \pm 0.86)\,\%$ with the ideal truth tables for the CPF and CNOT gates, respectively, which is in good agreement with the simulations. Using the gate to entangle the atomic and photonic qubits, I further measure a Bell state fidelity of $(65.90 \pm 1.73)\,\%$, exceeding the classical threshold of $50\,\%$ and confirming the generation of atom-photon entanglement. This represents a key building block for future quantum network protocols.

Contents

Chapter 1

Introduction

Few theories have reshaped modern technologies and thus our day to day life as profoundly as quantum mechanics. Its origins trace back to the early 1900s, when Max Planck's analysis of black body radiation led to the formulation of Planck's law, describing the quantized exchange of energy between matter and the radiation field. This sparked the beginning of modern quantum mechanics from which a wide range of technologies has since emerged. Among the most consequential is the laser, whose principle of stimulated emission was first described by Albert Einstein [1] and which today underpins experimental techniques central to this work, including Sisyphus cooling [2] and the magneto-optical trap [3]. Another technology of great importance is the atomic clock, based on Norman Ramsey's work [4], enabling the precise definition of the International System of Units second [5]. One piece of technology, which shapes our modern civilization the most, is that of the transistor, specifically the metal–oxide–semiconductor field-effect transistor (MOSFET). The MOSFET transistor, whose operation is itself governed by the quantum mechanical band structure of semiconductors [6], is the most used type of transistor found in modern electronics [7] that either allows or denies the passage of electrical current. By either blocking the current or letting it pass through the component, one is able to construct a miniaturized switch, that can be used to realize a single *bit*.

In 1965, Gordon Moore extrapolated from a handful of data points to predict that the number of components on an integrated circuit would double with every year for the next decade [8]; in 1975, he revised this estimate to a doubling every two years. This projection, which came to be known as *Moore's Law*, was adopted by the semiconductor industry as a benchmark, turning the prediction into a self-fulfilling prophecy that guided chip development for more than half a century [9].

The drive toward ever-smaller components was foreshadowed by Richard Feynman's 1959 lecture *There's Plenty of Room at the Bottom* [10], which envisioned the storage of information at the atomic scale. However, the continuous shrinkage of transistors, and with it the exponential growth in computing performance, must eventually hit fundamental limits. As the individual chip components become smaller and approach the atomic scale, effects such as quantum tunneling and heat dissipation diminish the expected performance increase [11].

As computational tasks become increasingly demanding – for instance, finding the most efficient delivery route for a logistics company [12] or accurately simulating quantum many-body systems [13, 14, 15] – the scalability of classical computation reaches its practical limits, motivating the search for fundamentally new computation platforms. As Feynman points out in his talk "Simulating Physics with Computers":

> "Nature isn't classical, dammit, and if you want to make a simulation of nature, you'd better make it quantum mechanical, and by golly it's a wonderful problem, because it doesn't look so easy."[13]

So while classical computers were still trying to realize the trend Moore predicted in 1965, the foundation for a new paradigm of computation were already being laid. In 1980, Paul Benioff proposed the first quantum mechanical model for a Turing Machine [16], demonstrating that computation could in principle be carried out using quantum mechanical processes. This new way of computing is called *quantum computing*, which instead of using fixed bits that can either be a 1 or a 0, uses the so called *qubits*. A qubit is a two-state quantum-mechanical system, whose state can be any *superposition* of $|0\rangle$ and $|1\rangle$, i.e. $|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle$, where α and β are both complex numbers that must fulfill: $|\alpha|^2 + |\beta|^2 = 1$. That is, until a measurement is performed and the state collapses to $|0\rangle$ or $|1\rangle$ with probabilities $|\alpha|^2$ and $|\beta|^2$ respectively. What makes qubits interesting is the relationship multiple qubits can have with one another. Specifically, qubits can be *entangled* with one another, thus not allowing for description of one qubit state, without the state of another qubit. By leveraging these two fundamental quantum mechanical principles (superposition and entanglement), many protocols have been proposed that, in theory, should outperform their classical counterparts. Be it in the realm of encryption where certain quantum algorithms now promise an exponential speedup [17], in quantum key distribution where the laws of quantum mechanics enable fundamentally secure communication [18], or in the transfer of quantum states between distant nodes via quantum teleportation [19].

However, as time went on, and different hardware platforms were investigated that could serve as a foundation for a quantum computer, one problem (along many others), became apparent: scalability. As one tries to increase the amount of qubits used in a quantum computer, being able to keep external noise and errors that each component produces at bay, becomes more and more of a challenge [20]. In order to be able to scale these new systems up, it was proposed that instead of building one large quantum computer (where one would try to keep errors at bay via error correction algorithms integrated into the fundamental structure of the computer [21]), one would instead split the computational power up into multiple smaller decentralized pieces [22, 23, 24]. Each of these pieces would then represent a single quantum node, that is able to create entanglement between a stationary qubit and a mediating qubit, transfer the state of the mediating qubit onto the stationary one, and also process information through the use of quantum logic gates. By linking many of these quantum nodes, one aims to overcome size-scaling and error-correction problems that would limit the size of machines for quantum processing [23, 25]. This network of nodes would then appropriately be named a *Quantum Network*. One way of realizing such a quantum network is via the strong coupling interaction of single photons and atoms in the setting of cavity quantum electrodynamics (*cQED*). Here, each quantum node would store a set number of stationary atoms in between two highly reflective mirrors, which are able to be individually controlled, and interlinked with one another through another set of quantum channels, realized by photons that travel in between the nodes through optical fibers.

One of these cavity systems is the *CavityX* system at the Max Planck Institute for Quantum Optics, which features a ^{87}Rb atom coupled to two crossed fiber cavities, hence the letter X in the name. In the past this setup has been used to achieve atom-photon entanglement with a fidelity of 87% [26], successfully store a photonic qubit onto an atomic qubit with a fidelity of 94.7%[27], thereby demonstrating the first two ingredients of a quantum node: the generation of entanglement between stationary and mediating qubits, and the transfer of a photonic state onto an atomic one. It is currently being developed to realize atom-atom entanglement between two nodes connected via a telecom fiber. This fiber runs from the Ludwig-Maximilians-Universität (LMU) in the Munich city center to the Max Planck Institute for Quantum Optics at the research campus in Garching, forming a connection spanning over 24 km [28]. What remains, however, is the third ingredient: the ability to process quantum information locally through deterministic quantum logic gates. Realizing such a gate operation on the CavityX platform is the central goal of this thesis.

Atom-photon quantum gates have already been demonstrated in a number of platforms. In the setting of neutral atoms coupled to optical cavities, Reiserer et al. demonstrated the first quantum gate between a flying optical photon and a single trapped atom [29], which was later extended to a photon-mediated gate between two atoms in a cavity [30] and a photon-photon gate mediated by a single atom [31]. Similar atom-photon interactions have been explored in nanophotonic structures [32] and with silicon-vacancy centers in diamond nanocavities [33], and recently distributed quantum computing was demonstrated between two trapped-ion modules linked by a photonic channel [34].

Building on these advances, this thesis realizes an atom-photon controlled-phase gate – between an atomic clock qubit and a polarization-encoded photonic qubit – in the CavityX platform, following the in-principle deterministic protocol proposed by Duan and Kimble [22]. The scheme is implemented using the birefringent Fabry-Pérot cavity in the CavityX platform, whose two spectrally separated polarization eigenmodes naturally provide the coupled and uncoupled reflection channels required by the protocol, without the need for external phase stabilization. In previous realizations of the same protocol in non-birefringent cavities [29], the polarization selectivity was instead obtained through the splitting of the Zeeman states, which required encoding the atomic qubit in magnetically sensitive states. In the present work, the birefringent cavity eliminates this constraint, enabling the atomic qubit to be encoded in the magnetic-field-insensitive clock states of the $5S_{1/2}$ ground-state manifold of ^{87}Rb. The clock states exhibit no first-order Zeeman shift and are therefore robust against ambient magnetic field fluctuations. Together with the previously demonstrated capabilities for storing and reemitting quantum information on the CavityX platform, the gate realized in this thesis contributes to the set of primitives – storing, processing, and reemitting of quantum information – required to operate CavityX as a fully functional node of a quantum network.

The structure of this thesis is as follows. Chapter 2 introduces the theoretical framework of cavity quantum electrodynamics, including the effects of cavity birefringence and the mechanism of atom-state-dependent cavity reflection that underpins the gate operation. Chapter 3 describes the experimental platform, covering the cavity system, the trapping and positioning of a single ^{87}Rb atom, and the preparation of the atomic qubit in the clock states. Chapter 4 details the photonic qubit path, addressing polarization-dependent losses and the input-output optical setup. Chapter 5 presents the main results: the characterization of the atom-photon gate performance through gate truth tables and the generation of Bell states. Finally, Chapter 6 provides a summary and outlook.

Chapter 2

Theory

This chapter introduces the theoretical framework needed to understand the key points of this thesis. First, a general introduction into cavity quantum electrodynamics will be given, alongside the intricacies in the setup that will modify the general theory. Afterwards the reflection from an atom-cavity system will be explored, and finally how it can be used to construct an atom-photon quantum logic gate.

2.1 Cavity Quantum Electrodynamics

The field of *Cavity Quantum Electrodynamics* explains the interaction between a resonator (in this case a Fabry-Perot cavity), and electronic states within matter systems with which the light inside the resonator interacts (in this case the D_2 transition of ^{87}Rb). By placing the atom between two highly-reflective mirrors forming the optical cavity, one increases the number of times a photonic field that is inside a cavity gets reflected back and forth between the two mirrors, enhancing the interaction cross-section compared to free space. The strength of this enhancement is captured by the single-atom cooperativity $C = g^2/(2\kappa\gamma)$, which must satisfy $C > 1$ for efficient light-matter interaction at the single photon level [35].

The theoretical model which explains this interaction in a fully quantum mechanical picture is called the *Jaynes-Cummings Model* [36] and was named after Edwin Jaynes and Fred Cummings who developed it in 1963. To introduce this model, we consider a three-level emitter (i.e. the atom), which is interacting with a single cavity optical mode.

The *Jaynes-Cummings* (JC) Model describes the coherent exchange of energy between a quantized cavity field and a two-level transition of the atom. The cavity supports a single optical mode at frequency ω_c. The atom, which is also located in the cavity, supports a transition frequency close to ω_c, from a coupled state $|c\rangle$ to an excited state $|e\rangle$ at a frequency ω_a. This transition is treated in the dipole approximation. In order to then complete the three-level-emitter picture relevant for the atom-photon gate, we include an uncoupled state $|u\rangle$. This uncoupled state, which lies *outside* the JC subspace, does not interact with the photon due to this state not being resonantly coupled to the cavity light mode. The inclusion of this extra state is what enables the conditional atom-photon gate operation.

Lastly, any real atom-cavity system, additionally suffers from losses, primarily the spontaneous decay of the atomic excitation at a rate of 2γ and the cavity field decay at a rate κ.

The full Hamiltonian which describes interaction between the atom and the cavity thus reads: [36]

$$\mathcal{H}_{\mathcal{JC}} = \underbrace{\hbar\omega_c \hat{a}^\dagger \hat{a}}_{\widehat{=}\text{cavity field}} + \underbrace{\hbar\omega_a \hat{\sigma}_{ce}^\dagger \hat{\sigma}_{ce}}_{\widehat{=}\text{atom}} + \underbrace{\hbar g(\hat{a}\hat{\sigma}_{ce}^\dagger + \hat{a}^\dagger \hat{\sigma}_{ce})}_{\widehat{=}\text{light-atom interaction}} \tag{2.1}$$

The three terms in (2.1) describe the energy of the cavity field, the energy of a bare atom, and the coherent coupling between the two governed by a coupling strength g.

The operators $\hat{a}^\dagger$ and $\hat{a}$ are the creation and annihilation operators of the cavity light modes, whereas $\hat{\sigma}_{ce}^\dagger = |e\rangle\langle c|$ and $\hat{\sigma}_{ce} = |c\rangle\langle e|$ are the atomic raising and lowering operators, respectively. The form of (2.1) assumes the so-called *Rotating-Wave Approximation* in which fast oscillating terms at optical frequencies can be neglected.

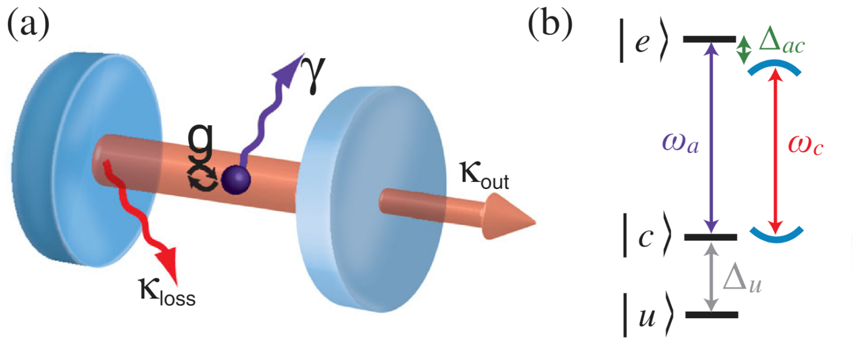


Figure 2.1: (a) Basic physical situation considered in cQED. A three-level emitter, e.g., a single atom, is located in an optical cavity, schematically represented by two facing mirrors. The atom exchanges energy with the optical field at a rate $2g$. The decay rate of photons out of the resonator is $2\kappa = 2\kappa_{\text{out}} + 2\kappa_{\text{loss}}$. The atomic polarization decay rate is γ (b) Atomic energy level scheme. The transition frequency between the coupled ground state $|c\rangle$ and the excited state $|e\rangle$ is ω_a. The cavity resonance frequency is ω_c. Atom and cavity are detuned by $\Delta_{ac} = \omega_a - \omega_c$. The third, uncoupled level $|u\rangle$ is detuned from $|c\rangle$ by Δ_u. Figure taken from [35].

Solving for the eigenenergies of the system one is left with:

$$E_0 = 0 \tag{2.2}$$

$$E_{\pm}(n) = n\hbar\omega_c + \hbar\frac{\Delta_{ac}}{2} \pm \frac{\hbar}{2}\sqrt{\Omega_n^2 + \Delta_{ac}^2}, \quad n \in \mathbb{N} \tag{2.3}$$

In (2.3), n is the number of excitation quanta in the system, with $\Omega_n = 2g\sqrt{n}$ being the corresponding n-quanta Rabi frequency. The term Δ_{ac} describes the detuning of the cavity resonance frequency from the atomic transition. The main take-away from (2.3) is that on resonance ($\omega_a = \omega_c$), the eigenenergies of the system are split by $2\hbar g\sqrt{n}$, giving rise to the so-called *Vacuum Rabi Splitting.*

2.1.1 Cavity Birefringence

Equation (2.1) only covers the case for having a single cavity light-mode. The setup used in this thesis, however, features a birefringent cavity. The birefringence is created by physically changing the geometric shape of the cavity mirror during production of the fiber ends.[37] This forces a change in the optical path lengths between different polarizations, leading to a frequency splitting $\Delta\nu$ proportional to the ellipticity ϵ of the mirror surfaces:[38]

$$\Delta\nu = \frac{\nu_{FSR}}{2\pi k}\frac{\epsilon^2}{R_2}. \tag{2.4}$$

The property ν_{FSR} in (2.4) describes the free spectral range, $k = \frac{2\pi}{\lambda}$ the photon wavenumber, and R_2 the radius of curvature of the cavity mirror.
This means that we need to consider a system with two different light modes for the JC Hamiltonian, which in this case are the π and the V modes. However only the π mode is resonantly coupled to the $|c\rangle \leftrightarrow |e\rangle$ transition, whereas the V mode is detuned by $\Delta_{\text{biref}} = 2\pi \cdot 500\text{MHz}$. In the context of this thesis $\Delta_{\text{biref}} \gg \kappa, g, \gamma$. This effectively makes the V-mode uncoupled, and can be neglected in the analytical treatment. It is, however, still included in the Hamiltonian for use in numerical simulations where both modes are driven explicitly. This leads to the extended Hamiltonian:

$$H_{JC} = \hbar\omega_{c,\pi}\hat{a}_\pi^\dagger\hat{a}_\pi + \hbar(\omega_{c,\pi} + \Delta_{\text{biref}})\hat{a}_V^\dagger\hat{a}_V + \hbar\omega_a\hat{\sigma}_{ce}^\dagger\hat{\sigma}_{ce} + \hbar g(\hat{a}_\pi\hat{\sigma}_{ce}^\dagger + \hat{a}_\pi^\dagger\hat{\sigma}_{ce} + \hat{a}_V\hat{\sigma}_{ce}^\dagger + \hat{a}_V^\dagger\hat{\sigma}_{ce}) \tag{2.5}$$

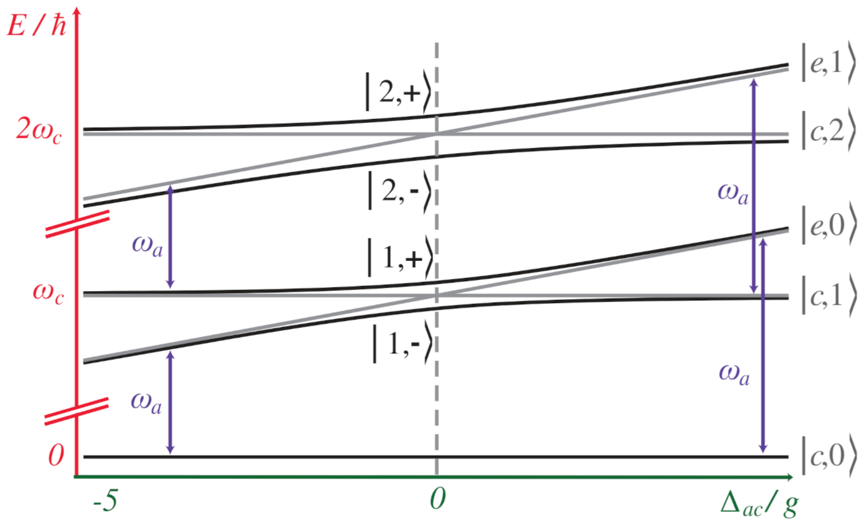


Figure 2.2: The Jaynes-Cummings ladder of the dressed energy eigenstates $|N, \pm\rangle$ of a strongly coupled atom-cavity system, where the cavity frequency is kept fixed and the emitter frequency is scanned over the resonance. Compared to the uncoupled situation (gray), the coupled states (black) exhibit pronounced avoided level crossings. At zero detuning, the levels are separated by $2g\sqrt{N}$. Figure taken from [35].

2.1.2 External Drive

At the heart of the atom-photon gate lies the reflection of the incoming photon off the cavity system. In order to describe this in a quantum mechanical way, one needs to include a driving term in the Hamiltonian with the form of $H_{D,S} = i\hbar(\epsilon e^{-i\omega_d t} + \epsilon^* e^{i\omega_d t})(\hat{a} + \hat{a}^\dagger)$. This Hamiltonian $H_{D,S}$ is defined in the Schrödinger picture, and has a complex drive amplitude ϵ and driving field frequency ω_d. After applying the rotating wave approximation (RWA), which neglects the fast oscillating terms in $H_{D,S}$, and changing into the rotating wave frame to remove the explicit time dependence, one gets the following expression:

$$H_D = \mathrm{i}\hbar \left(\hat{a}_s \epsilon^* + \hat{a}_s^\dagger \epsilon\right) \tag{2.6}$$

where $\hat{a}_s$ denotes the annihilation operator of the cavity mode matching the input photon polarization (s $= \pi$ or V).
Adding (2.6) to (2.5), one gets the following full Hamiltonian:

$$\begin{aligned} H_{JC} = &\hbar\Delta_{c,\pi}\hat{a}_\pi^\dagger\hat{a}_\pi + \hbar\left(\Delta_{c,\pi} + \Delta_{\text{biref}}\right)\hat{a}_V^\dagger\hat{a}_V + \hbar\Delta_a\hat{\sigma}_{ce}^\dagger\hat{\sigma}_{ce} \\ &+ \hbar g\left(\hat{a}_\pi\hat{\sigma}_{ce}^\dagger + \hat{a}_\pi^\dagger\hat{\sigma}_{ce} + \hat{a}_V\hat{\sigma}_{ce}^\dagger + \hat{a}_V^\dagger\hat{\sigma}_{ce}\right) \\ &+ \mathrm{i}\hbar\left(\hat{a}_s\epsilon^* + \hat{a}_s^\dagger\epsilon\right) \end{aligned} \tag{2.7}$$

This then gives a definition of the JC Hamiltonian with respect to an external driving field, and based on the detunings of the driving field with respect to the cavity resonance frequency $\Delta_{c,\pi} = \omega_{c,\pi} - \omega_d$ and the atomic transition frequency $\Delta_a = \omega_a - \omega_d$, respectively.

2.2 Reflection from an Atom-Cavity System

For quantum information processing with photonic qubits, the cavity must be designed so that photons enter and leave through a single well-defined external mode. This is achieved by making one mirror highly transmissive — the outcoupling (OC) mirror — while keeping the opposing mirror highly reflective (HR). Photons then enter and exit predominantly through the OC mirror. The total cavity decay rate decomposes into contributions from each loss channel:

$$\kappa = \kappa_{oc} + \kappa_{HR} + \kappa_{paras} \tag{2.8}$$

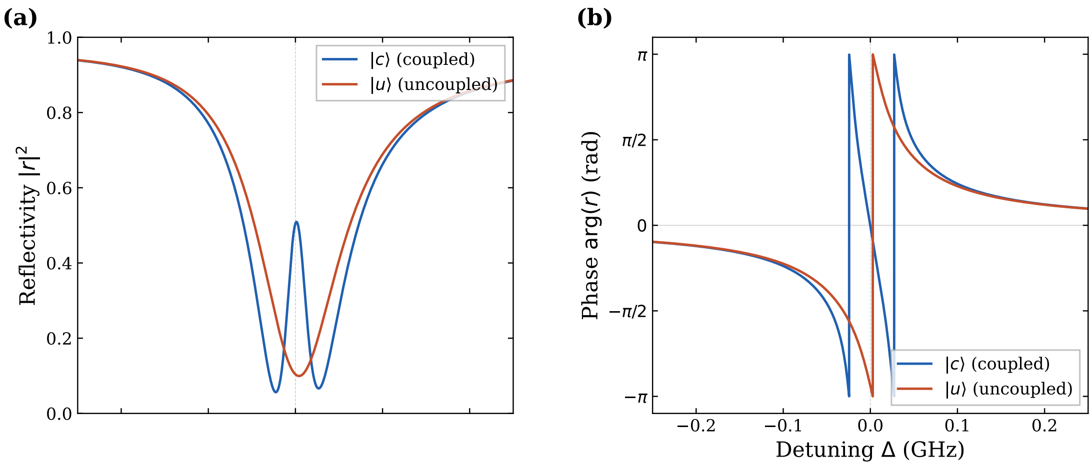


Figure 2.3: Reflectivity $|r|^2$ (a) and phase $\arg(r)$ (b) of the reflection coefficient as a function of detuning Δ between the probe light and the cavity resonance, for the coupled state $|c\rangle$ (blue) and the uncoupled state $|u\rangle$ (red). The coupled state exhibits vacuum Rabi splitting, visible as the double-dip structure in reflectivity and the associated phase features around resonance. At zero detuning, the uncoupled state $|u\rangle$ acquires a π phase shift, whereas the coupled state $|c\rangle$ does not. Parameters used are $(g,\, \kappa,\, \gamma)/2\pi = (26,\, 58,\, 3.0)\,\text{MHz}$, with output-coupler fraction $\kappa_{\text{oc}}/\kappa = 0.85$ and mode matching parameters $\mu_{\text{rf}} = 0.978$, $\mu_{\text{fc}} = 0.873$.

In (2.8), κ_{oc} and κ_{HR} describe the rate at which light couples into and out of the cavity through the OC and HR mirror respectively, whereas κ_{paras} describes the rate of losses due to absorption at the mirror surfaces. For single-sided operations one needs to ensure that $\kappa_{oc} \gg \kappa_{HR}, \kappa_{paras}$. The escape probability of the photon is then defined by the outcoupling ratio $\frac{\kappa_{oc}}{\kappa}$.

Another important quantity is the matching of the spatial modes of the light when coupling into the cavity. As the incoming light gets coupled into the cavity, a mismatch between the spatial modes in transmission and reflection can occur. This lowers the efficiency at which the light can couple to the cavity itself. The mode matching is then given by the overlap integral of the mode functions for the light one is investigating. In this case it is important to quantify the mode matching between the reflected light at the first cavity and the incoming light (μ_{fr}) as well as the one for the incoming light and the optical mode inside of the cavity (μ_{fc}).

The field incident on the cavity is treated as an input field. This then can be linked to an output field via input-output formalism. When driving the system only from one side, as it is the case here, the relation between input and output becomes:[39]

$$\hat{a}_{\text{out}}(t) = \mu_{fr}\, \hat{a}_{\text{in}}(t) + \mu_{fc}\sqrt{\kappa_{oc}}\, \hat{a}(t) \tag{2.9}$$

Here (2.9) was adjusted to also consider the mode matching parameters. Solving Langevin equations for weak coherent driving and photonic wave envelopes which vary only on a time scale longer than κ, one can write down the analytical representation of the reflection amplitude[40, 32, 41]:

$$r(\omega) = \mu_{fr} - \mu_{fc}^2 \frac{2\kappa_{oc}(i\Delta_a + \gamma)}{(i\Delta_{c,\pi} + \kappa)(i\Delta_a + \gamma) + g^2} \tag{2.10}$$

The reflection amplitude in (2.10) depends on the detuning $\Delta_{c,\pi} = \omega - \omega_{c,\pi}$ between the driving laser frequency and the cavity resonance frequency $\omega_{c,\pi}$ (for the π mode) and on the detuning relative to the atomic transition frequency $\Delta_a = \omega - \omega_a$.

2.3 Atom-State-Dependent Cavity Reflection

Looking at (2.10) one can see that the behavior of the reflection amplitude at zero detuning ($\Delta_a = \Delta_{c,\pi} = 0$) is:

$$r(\Delta = 0) = \mu_{fr} - \mu_{fc}^2 \frac{2\kappa_{oc}\gamma}{\kappa\gamma + g^2} \tag{2.11}$$

In the event of an uncoupled atom-cavity system (effectively $g = 0$), which would be achieved by preparing the atom in the $|u\rangle$ state, the reflection coefficient $r < 0$. This would imply a relative phase shift of π between the reflected and incoming light field ($a_{out}(t) \approx -a_{in}(t)$). Preparing the atom in the coupled state $|c\rangle$ however, the Vacuum Rabi Splitting shifts the resonances of the atom-cavity system to $\omega_c \pm g$, so the cavity appears off-resonant for a photon at ω_c. The photon is therefore reflected from the OC mirror without entering the cavity, acquiring no phase shift, which is mirrored by a reflection coefficient of $r > 0$ ($a_{out}(t) \approx a_{in}(t)$). This behavior is shown in Figure 2.3.

A V polarized photon, being far detuned from the cavity resonance by Δ_{biref}, is reflected without entering the cavity and acquires no phase shift, regardless of the atomic state. This state-dependent phase shift of the output field depending on the state of the atom, enables the realisation of a quantum atom-photon gate with the atom being the controlling qubit and the photon being the target qubit.[22]

Adopting standard quantum computing notation, the uncoupled state of the atom is the $|0\rangle$ state, and the coupled one the $|1\rangle$ state. The photon, either π-polarized or V-polarized, is represented as the input states $|\pi\rangle$ and $|V\rangle$. Furthermore, for all input states, the input drive is resonant with the π mode of the birefringent cavity.

For an input state $|0, \pi\rangle$, meaning the atom is prepared in the uncoupled state $|0\rangle$ and a π-polarized photon is sent in, the photon will be able to enter the cavity and (ideally) exit through the OC mirror, with a π phase-shift. For the input state $|1, \pi\rangle$, the Vacuum Rabi splitting shifts the cavity resonance away from the photon frequency, causing it to be reflected without a phase shift. When sending a $|V\rangle$ photon, it will get reflected off of the OC mirror regardless, due to the birefringence of the cavity. This results in the following reflections:

$$\begin{aligned} |0,\pi\rangle &\rightarrow -|0,\pi\rangle \\ |0,V\rangle &\rightarrow |0,V\rangle \\ |1,\pi\rangle &\rightarrow |1,\pi\rangle \\ |1,V\rangle &\rightarrow |1,V\rangle \end{aligned} \tag{2.12}$$

This output from the atom-photon reflection gives us a *Controlled Phase Flip* (CPF) gate. This conditional phase shift allows the construction of a universal quantum gate that can be transformed into any two-qubit gate using rotations of the individual qubits, which in the case of the photon can be done via retardation waveplates.

Rotating the photonic basis states to:

$$\begin{aligned} |+\rangle &= \frac{|V\rangle + |\pi\rangle}{\sqrt{2}} \\ |-\rangle &= \frac{|V\rangle - |\pi\rangle}{\sqrt{2}} \end{aligned} \tag{2.13}$$

and keeping the atom either in the coupled ($|1\rangle$) or uncoupled case ($|0\rangle$), results in the following reflection:

$$\begin{aligned} |0,+\rangle &\rightarrow |0,-\rangle \\ |0,-\rangle &\rightarrow |0,+\rangle \\ |1,+\rangle &\rightarrow |1,+\rangle \\ |1,-\rangle &\rightarrow |1,-\rangle \end{aligned} \tag{2.14}$$

In the new photonic basis $\{|+\rangle, |-\rangle\}$, the CPF gate is equivalent to a *Controlled-NOT* (CNOT) gate, with the atom as the controlling qubit and the photon polarization as the target qubit. The photon polarization is flipped ($|+\rangle \leftrightarrow |-\rangle$) when the atom is in $|0\rangle$, and left unchanged when the atom is in $|1\rangle$.

Chapter 3

Atom-Cavity platform

The system uses a single ^{87}Rb atom coupled to a birefringent Fabry-Perot cavity inside an ultra-high vacuum (UHV) chamber. The optical beam path used for the realization of the atom-photon gate is part of a much larger setup. The full setup, named *CavityX*, consists of two Fabry-Perot cavities arranged in an X formation. For the atom-photon gate, parts of the setup like the magneto-optical trap, the 850 nm red-detuned trap laser, and the microwave antennas integrated into the UHV chamber are crucial. Other components, most notably the second cavity, are not within the scope of this thesis, but will be discussed in the outlook.

3.1 Cavity system

At the heart of the CavityX system are the two crossed fiber cavities which are named the "short" cavity (SC) and the "long" cavity (LC). The two cavities differ in two ways. First, the short cavity is physically shorter than the long one (79 µm vs. 161 µm). Second, the long cavity was designed to minimize birefringence, while the short cavity was designed to maximize it. Each cavity fiber is held in place by an in-house built holder shown in figure 3.1.
Each cavity is formed by two opposing fibers — one single-mode (SM) and one multi-mode (MM) — as shown in Figure 3.1(b). The incoming light is coupled in and out via the SM fiber. In order to ensure that most of the coupled light exits through the SM fiber, different mirror coatings are used for the fiber surfaces. The MM fiber surfaces are coated with a highly reflective dielectric coating, while the opposing SM fiber has a dielectric coating with a lower reflectivity (10 ppm transmission vs. 340 ppm transmission). This asymmetric design ensures the cavity is single-sided, causing most of the coupled-in light to then exit the cavity through the same fiber through which it entered the system [42].
The cavity fibers are fabricated using a CO_2 laser machining technique by a previous member of the Quantum Dynamics division at MPQ. The process of machining technique is described in Manuel Brekenfeld's thesis [37]. Changing the beam profile of the CO_2 laser allows for the fabrication of elliptical surfaces, as is the case for the short cavity. This leads to a cavity mode splitting, i.e., cavity birefringence, as shown in figure 3.2

3.2 Trapping of a Single Atom

3.2.1 Magneto-optical trap

In order to realize an atom-photon quantum gate, one is required to have an atom trapped inside the cavity with which the photon can then interact. Therefore one needs to properly position the atom and keep it trapped at the position where it most efficiently couples to the cavity to ensure the most efficient gate operation.

The starting point for this is the trapping of an atomic cloud via a magneto-optical trap (MOT). Once the atomic cloud is trapped, one turns off the MOT, causing the atoms to fall down due to

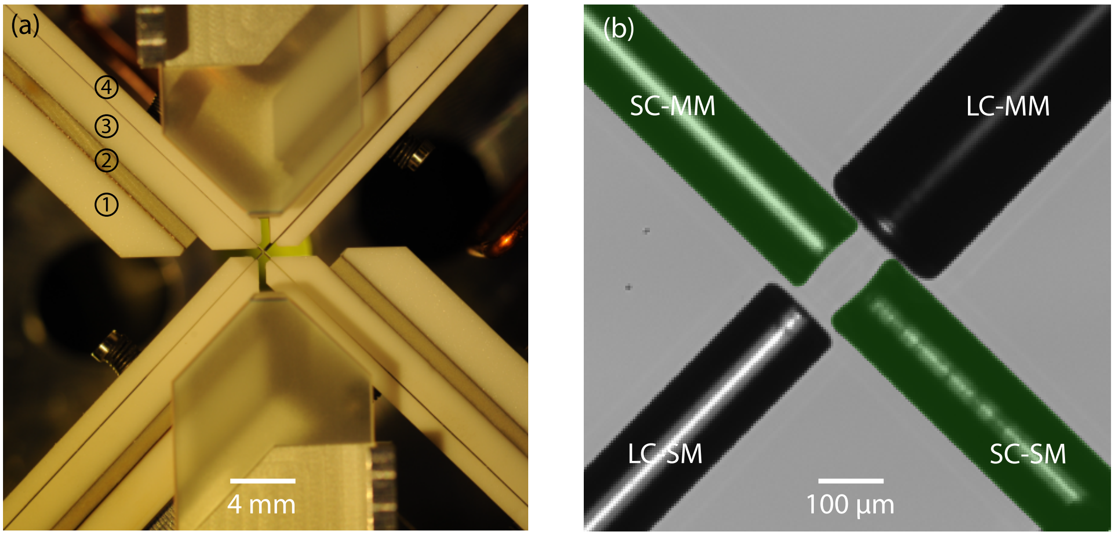


Figure 3.1: Vertical top view of the construction holding the crossed fiber cavities in place. The holder of each fiber consists of 4 elements, each of which is labeled in the picture: ① Base element for mounting the fiber holder to downstream components; ② Shear piezo actuator for active stabilization of the resonator length; ③ Mount in which a V-groove is fabricated - where the fiber is positioned and mechanically clamped; ④ Lid to clamp the fiber within the V-groove mount. A zoom of the center of the picture is shown in (b). The green-colored cavity is 75.7 μm long and is composed of a single- (SC-SM) and multi- (SC-MM) mode fiber with reflective coatings analogous to LC-SM and LC-MM, respectively. The black-colored cavity is the long cavity, but since it is not used in the context of this thesis it will not be discussed further. This photo has been colored in post-production. Figure taken from [42].

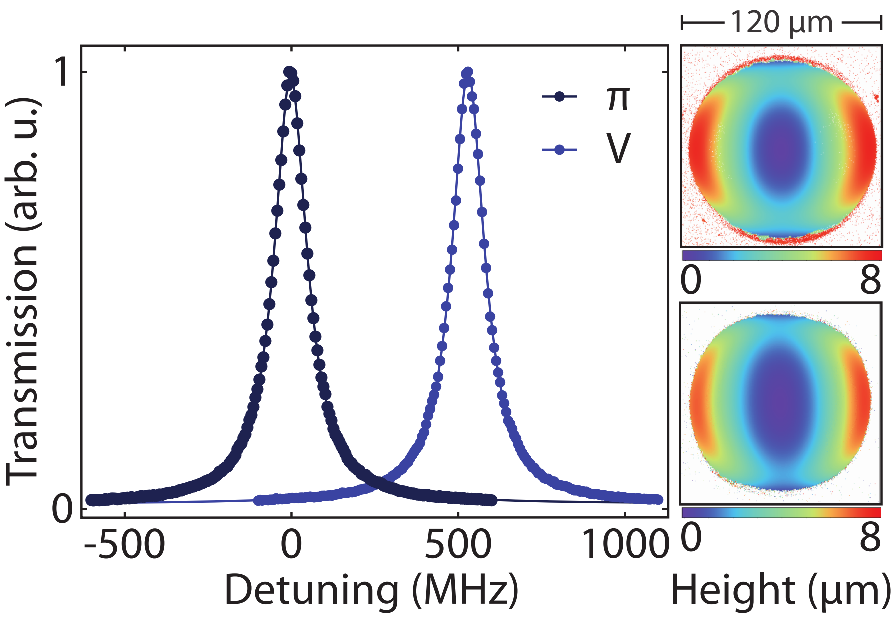


Figure 3.2: Transmission spectrum was recorded for the short cavity. The short cavity exhibited a large polarization mode splitting, a result of the strong ellipticity of its mirror surfaces. The error bars are smaller than the data point markers. Figure taken from [37].

Parameter	Short Cavity
Radius of curvature of OC_1	$115\,\mu$m and $283\,\mu$m
Radius of curvature of HR_2	$96\,\mu$m and $167\,\mu$m
Cavity length	$75.7\,\mu$m
Free spectral range	1.98043 THz
Waist radius	$3.5\,\mu$m and $4.4\,\mu$m
Waist radius pos. from OC	$25.8\,\mu$m and $22.7\,\mu$m
Mode waist at OC	$3.9\,\mu$m and $4.6\,\mu$m
Mode waist at HR	$5.9\,\mu$m and $5.2\,\mu$m
Mode waist at cavity center	$3.5\,\mu$m and $4.4\,\mu$m
Mode volume ($\lambda = 780\,$nm)	$1900 \cdot \lambda^3$
Mode matching	0.873
$g(1' \leftrightarrow 2)/(2\pi)$ at cavity center	25.9(5) MHz
Transmission through OC	340 ppm
Transmission through HR	10 ppm
Parasitic losses	50 ppm
Total losses	400 ppm
$\kappa/(2\pi)$	58 MHz
Linewidth	116 MHz
Finesse	16531
Outcoupling ratio κ_{oc}/κ	0.85
Eigenmodes freq. splitting	505.0(5) MHz

Table 3.1: Cavity parameters.

the gravitational force, from which single atoms can then be loaded into the cavity's center. The concepts of a MOT are explained in [43].
In order to trap atomic clouds via a MOT, one needs to generate a rubidium vapor inside the UHV chamber. This is done via light-induced atom desorption (LIAD), in which atoms are dispensed from a rubidium dispenser that is placed inside of the vacuum chamber. The light source is a UV diode ILH-XQ01-S410 with an emitting wavelength of 410nm, which is pulsed for the length of time needed to load the MOT.

The MOT loading procedure can be divided into two phases: First comes the loading phase in which the MOT coils, the MOT beams (a cooling laser and a repumper laser), and the UV light are switched on simultaneously. This then creates the atomic cloud and cools it down to near the Doppler temperature of $146\,\mu$K for the ^{87}Rb D_2-line. [44]
Next an optical molasses is applied, which aims at bringing the atom to a temperature much below the Doppler limit. Here the detuning of the MOT beams is increased, while the optical power is decreased.
At the end of the MOT cycle, the MOT coils, beams and the UV light are switched off.

3.2.2 Trapping and Positioning of Single Atom

At the end of the MOT loading phase, the ultra-cold atomic cloud then sits approximately 10 mm above the cavity plane. Turning off the MOT causes the atoms to fall towards the cavity due to gravity, until they reach the cavities' crossing point, at which point cooling and trapping laser beams are switched on.
The cooling and repumping laser are shone diagonally onto the crossing point in a counter-propagating configuration. Using orthogonal polarizations in a lin$\perp$lin configuration for the counter-propagating lasers, the atoms are cooled to sub-Doppler temperatures via Sisyphus-cooling.
Once the atom is at the center of the cavity, two lasers are responsible for keeping it trapped there:
The first one is the intracavity laser of the short cavity which has a wavelength of 772 nm, and is blue-detuned from the D_2-line. The laser self-interferes in the cavity, forming a standing wave, thus confining the atom at the nodes, i.e., interference minima. Stabilizing the cavity length is

Table 3.2: Parameters used for the MOT loading phase.

MOT coil current	23.5 A
UV diode current	140 mA
MOT beams' diameter	12 mm
Loading phase	
Duration	1 s
Cooling laser detuning from $5S_{1/2}, F = 2 \leftrightarrow 5P_{3/2}, F' = 3$	-13 MHz
Repumping laser detuning from $5S_{1/2}, F = 1 \leftrightarrow 5P_{3/2}, F' = 2$	0 MHz
Cooling laser power	20 mW
Repumping laser power	250 μW
Molasses phase	
Duration	1 ms
Cooling laser detuning from $5S_{1/2}, F = 2 \leftrightarrow 5P_{3/2}, F' = 3$	-31 MHz
Repumping laser detuning from $5S_{1/2}, F = 1 \leftrightarrow 5P_{3/2}, F' = 2$	-20 MHz
Cooling laser ramp-down power	1.7 mW
Repumping laser power	250 μW

Parameters used for the MOT loading phase. The UV diode and the MOT coils are permanently on the whole time. The cooling and repumping laser power is measured behind a polarizing beam splitter, which separates the horizontal MOT beam from the top-diagonal ones.

achieved by first locking the intracavity laser using the PDH-lock technique and then locking the laser frequency to an optical frequency comb, which serves as a stable frequency reference. For the atom-photon gate the frequency of the laser was chosen so that the cavity is resonant to the D_2-line at 780 nm.

The second laser is the 852 nm vertical trap laser, that's shone in from the above the cavities' crossing point. The laser is linearly polarized along the long cavity axis, and focussed to a diameter of about 20 μm. It is then transmitted through the chamber and collimated to a mode diameter of 1cm. Lastly it gets transmitted through a dichroic element, a set of retardation waveplates, a piece of silica glass mounted to a galvanometer scanner (galvo), to then get retroreflected in a cat-eye configuration. Due to the laser being far detuned from the D_1 and D_2 lines of the atom, it forms a trapping standing wave, where the atoms are confined in the antinodes, i.e., interference maxima.

In order to achieve maximum cavity coupling, the galvo scans its amplitude, rotating the silica glass and thereby modifying the beam's angle of refraction, which shifts the standing-wave antinodes vertically. While this is happening, the fluorescence counts coming from the atom during the cooling phase are collected via the short cavity and detected with superconducting nanowire single-photon detectors (SNSPDs). The fluorescence counts are monitored in real time by an FPGA, which also controls the galvo scan amplitude. One can use this determine the height and thus the maximum coupling position of the atom from the fluorescence counts. In order to ensure that only a single atom was trapped instead of two weakly coupled atoms, the FPGA sequence is as follows:

1. The FPGA starts scanning the voltage of the galvo while monitoring the amount of fluorescence counts detected in a 25 ms window coming from the atom.
2. If the fluorescence counts go above a certain threshold (noise-threshold), this means that an atom that is weakly coupled to the cavity is emitting fluorescence counts into the cavity.
3. The FPGA then continues increasing the voltage, keeping track of the voltage (i.e., position) at which the maximum fluorescence counts were reached. In the meantime the clicks at each galvo step are summed into a single 32-bit register. This effectively represents an integral over the fluorescence counts as a function of the galvo amplitude curve.
4. Once the fluorescence counts drop below the (noise-threshold), meaning that the atom has been moved out of the cavity center, the integral of the fluorescence counts is evaluated. If

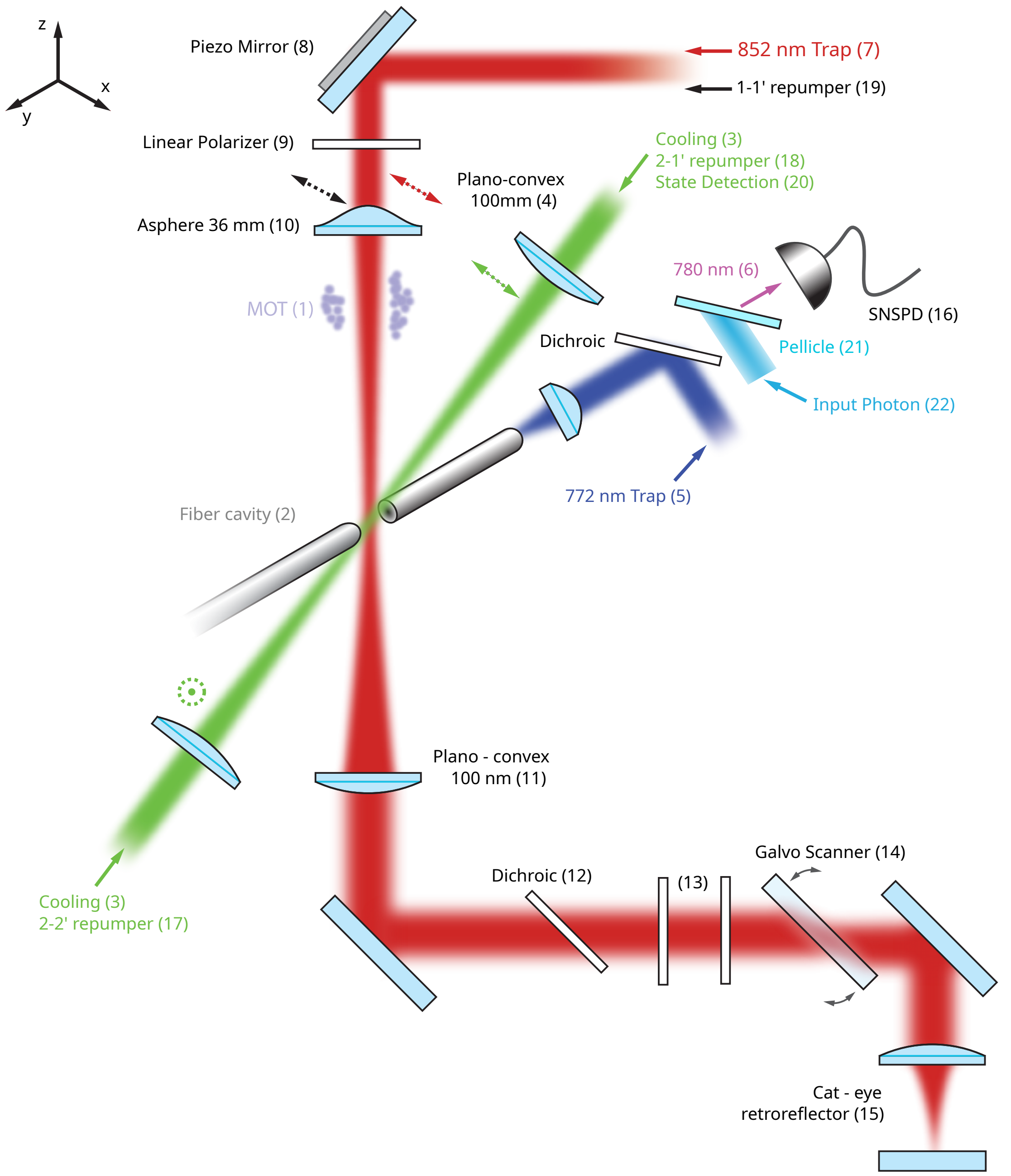


Figure 3.3: Overview of experimental setup used in this work. According to the reference frame, the z, the x, and the y axes are, respectively, the vertical trap, the long Cavity (not shown here), and the short Cavity axes. The dashed arrows and the dashed circle indicate the laser beams' polarization.

the integral is within a certain range, the FPGA will set the voltage at which maximum fluorescence counts were detected. If the integral is out of range (too low: atom was badly coupled anyway; too high: multiple atoms were in one node), the galvo keeps scanning until it either finds a new atom or reaches the maximum amplitude, causing the MOT and single-atom positioning loop to start again.

Figure 3.4 gives a visual overview of the trace during the positioning sequence.

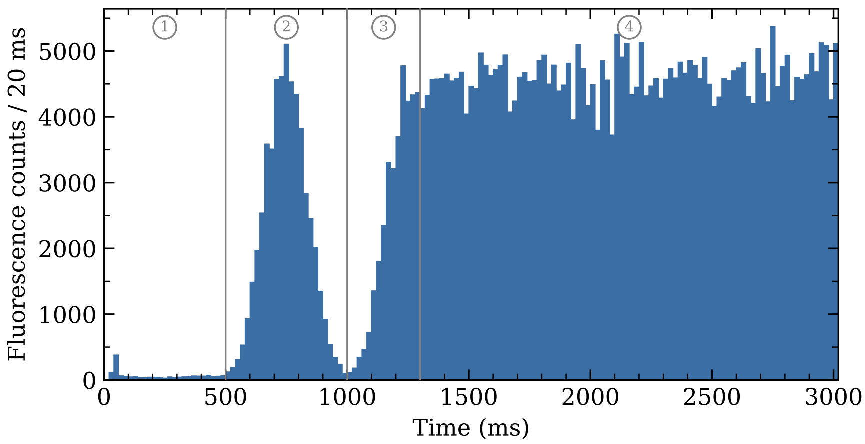


Figure 3.4: Typical trace in which fluorescence counts are recorded during the atom positioning procedure as a function of time. The MOT release sets the time = 0s. Around time 0.1s, an atom is trapped in the 852 nm trap and is not coupled to the cavity (1). Around time 0.5 s, the positioning procedure begins, in which the galvo is swept in one direction. During this time, the counts are recorded (2). Once the maximum coupling position is reached (3), the positioning sequence is interrupted, and the atom is kept in place while being cooled (4).

3.3 Atomic Qubit Preparation

For the atom-photon gate, the single atoms must be prepared in the two $m_F = 0$ ground states of the D_2-line, those being $5S_{1/2}, F = 1, m_F = 0$ for the uncoupled state, and $5S_{1/2}, F = 2, m_F = 0$ for the coupled state.
The first step in achieving this is to optically pump (OP) the atom into the desired state. This is done by repeatedly exciting the atom from a level one wishes to depopulate to an excited state, from which it will then decay into various ground states, determined by the branching ratios of the respective transition. In order to then populate the desired state, one uses a frequency for the pump laser, where all ground states one wants to depopulate are coupled to the excited state, except for the desired one, gradually populating the target state over time.
In this work, one first pumps the atom into the $F = 1, m_F = 0$ state. This is done by using two laser beams, one that is resonant with the $F = 2 \rightarrow F' = 2$ transition of the D_2-line of ^{87}Rb, which will lead to the depopulation of the $F = 2$ manifold and population of the $F = 1$ manifold. In order to then populate only the $F = 1, m_F = 0$ state, one applies another beam, resonant to the $F = 1 \rightarrow F' = 1$ transition of the D_2-line. Furthermore the beam must be π-polarized, since the $F = 1, m_F = 0 \rightarrow F' = 1, m_F = 0$ transition is forbidden [44]. The transition scheme can be seen in figure 3.5.
Using this technique one is able to populate the $F = 1, m_F = 0$ level with a probability of $\geq 97\,\%$.[42]

For the atom-photon gate, one also needs to be able to prepare the $5S_{1/2}, F = 2, m_F = 0$ for the coupled state. This is achieved by first preparing the atom in the already prepared $F = 1, m_F = 0$ state and using a set of microwave antennas integrated into the UHV chamber, to drive the $|F = 1, m_F = 0\rangle \leftrightarrow |F = 2, m_F = 0\rangle$ clock transition.
Finding the transition frequency is done by performing a microwave spectroscopy measurement. The atom is pumped into the $F = 1, m_F = 0$ via the previously discussed technique. By scanning

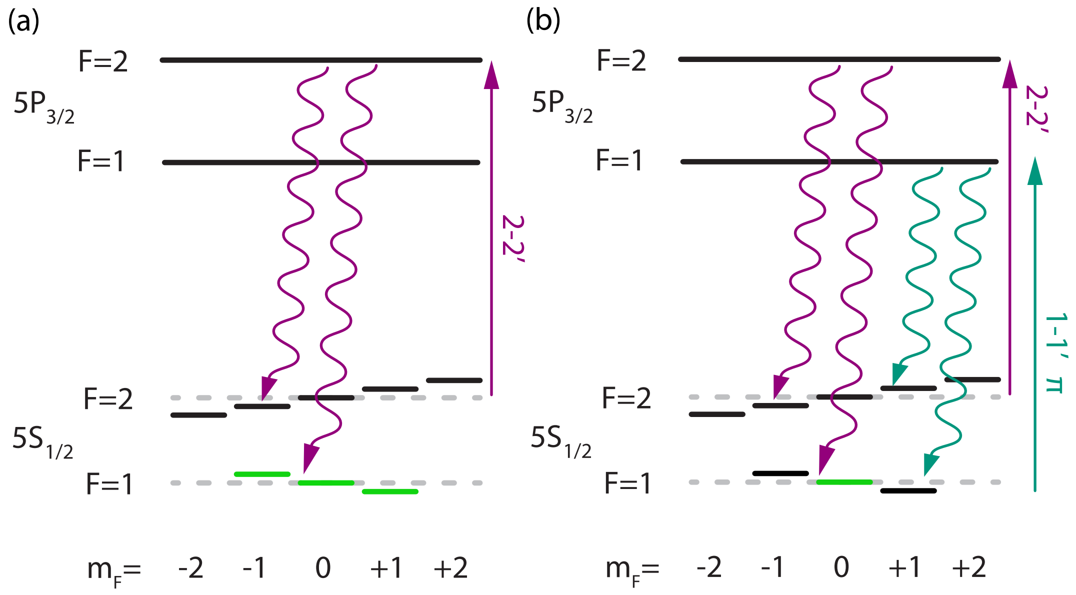


Figure 3.5: Schemes utilized for OP in $F = 1$ (a), and $F = 1, m_F = 0$ (b). Each sketch shows the used lasers (straight arrows), the atomic decay (wavy arrows), and the targeted state (light green lines).

the microwave frequency at fixed phase, one measures the F=2 population via cavity-enhanced fluorescence state detection (SD) with a detection window of 7.5 μs.
The resulting MW spectrum shown in 3.6 then shows three peaks, which correspond to the $F = 1, m_F = 0 \leftrightarrow F = 2, m_F = 0$ and $F = 1, m_F = 0 \leftrightarrow F = 2, m_F = \pm 1$ transitions. From this one can infer that the optical pumping into the $F = 1, m_F = 0$ is done with an efficiency of 97(1)%.[42]

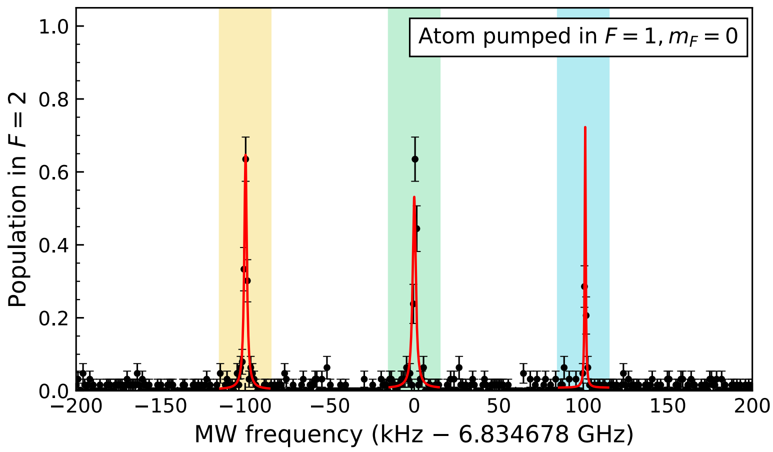


Figure 3.6: Resulting MW spectrum when the atom is initialized in $F = 1, m_F = 0$. The yellow, green, and blue colored bands mark the transitions from $F = 1, m_F = 0$ to $F = 2, m_F = -1$, $F = 2, m_F = 0$, and $F = 2, m_F = +1$, respectively. The red solid lines are Lorentzian fits to each of the three peaks, from which the transition frequencies and linewidths are extracted.

Once the frequency for the $F = 1, m_F = 0 \leftrightarrow F = 2, m_F = 0$ transition is known, one can coherently manipulate the atomic population in the ground state. By tuning the MW on resonance, one can drive Rabi oscillations [45] between two ground states, where the population evolution in the Zeeman state of $F = 2$ can be modeled as:[46]

$$P_2 = P_1 \cdot \frac{1}{2}(1 - \cos(2\pi\Omega t) \cdot e^{-\chi t}) \tag{3.1}$$

where P_1 is the initial population after optical pumping in the state $F = 1, m_F = 0$, Ω is the

previously determined Rabi frequency, and χ is the damping rate. The result is a combination of a sinusoidal term which describes the coherent state transfer between the two levels, and an exponential factor representing the dissipative mechanism which damps the oscillations.
By scanning the length of the π pulse at a constant MW frequency, one can determine the ideal time it takes for a state to coherently transfer from $F = 1, m_F = 0$ to $F = 2, m_F = 0$. Figure 3.7 shows the resulting Rabi oscillations for an atom prepared in $F = 1, m_F = 0$ and transferred to the $F = 2, m_F = 0$ state. Using a drive with a frequency of 30.7 MHz and a length of $(37.07 \pm 0.05)\,\mu$s, one is able to prepare the atom in the $F = 2, m_F = 0$ state with a probability of $\sim 88\%$.

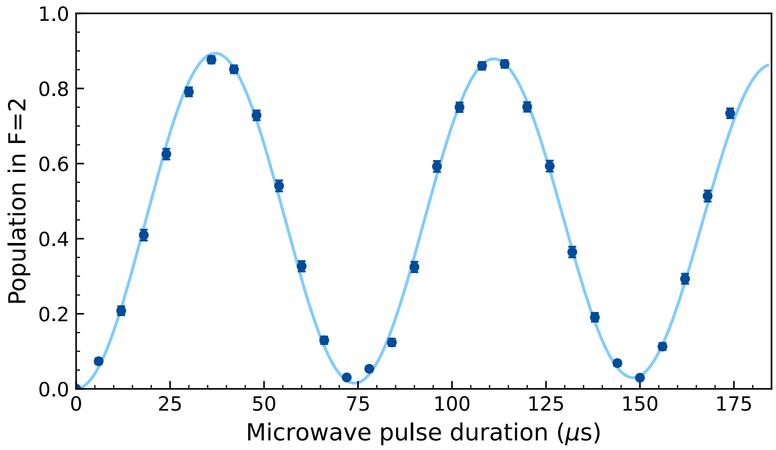


Figure 3.7: Rabi oscillations for the $F = 1, m_F = 0 \leftrightarrow F = 2, m_F = 0$ transition. The solid line represents the fit given by (3.1) to the data points.

Chapter 4

Photonic Qubit Path

This chapter describes the optical beam path used to deliver photonic qubits to the atom-cavity system and analyze them after reflection. The central challenge is that the birefringent cavity introduces a polarization-dependent amplitude mismatch of the reflected field requiring external compensation to achieve the optimal CNOT gate overlap.

4.1 Polarization-Dependent Losses

The reflection amplitude of a photon incident on an atom-cavity system, given by equation (2.10), depends sensitively on the system parameters: the coupling strength g, the cavity decay rate of the photons κ, the cavity outcoupling ratio κ_{OC}, the mode-matching between the fiber mode and the incoupling mirror surface μ_{fr}, the mode matching between the fiber mode and the cavity mode μ_{fc}, and the birefringence of the cavity $\Delta_{\text{biref}} = 2\pi \cdot 500\text{MHz}$. Looking at the reflection scheme for (2.14) in the basis defined in (2.13), it becomes apparent that in order to achieve the highest possible overlap with an ideal CNOT gate, the reflection amplitudes $|r_\pi|$ and $|r_V|$, must (ideally) be the same. However, for the cavity parameters realized in this system (characterized in section 5.1), this condition is not fulfilled.

As mentioned before, the V mode of the birefringent cavity is detuned by $\Delta_{\text{biref}} = 2\pi \cdot 500\text{MHz}$ from the π-mode, to which the atomic transition is resonantly coupled. This leads to a V polarized photon, getting reflected from the first OC mirror without entering the cavity due to being far off-resonant. This then leads to the V polarized component of both the CNOT input basis states, having a much larger relative reflection amplitude when compared to the π polarized portion.

A π-polarized photon, on the other hand, interacts with the atom inside the cavity, making the reflection amplitude dependent on the atomic state. In the coupled case $|1\rangle$, the vacuum Rabi splitting shifts the dressed-state resonances to approximately $\omega_c \pm g$. This then leads to the photon being reflected off the OC mirror, but not perfectly, due to finite cooperativity. In the uncoupled state $|0\rangle$, the π-polarized photon fully enters the cavity, undergoes many round trips, and transmits back out the OC mirror with a relative phase shift of π compared to the input field, on a timescale set by $1/\kappa_{OC}$. But due to intracavity losses (absorption, HR mirror leakage, imperfect mode-matching), the reflection amplitude is smaller than that of the coupled case.

Using the time-domain simulation described in Appendix A with the parameters characterized in section 5.1, the following reflection amplitudes are obtained:

	$\|0, \pi\rangle$	$\|1, \pi\rangle$	$\|0, V\rangle$	$\|1, V\rangle$
r	$-0.256 + 0.000i$	$0.704 + 0.031i$	$0.974 + 0.071i$	$0.973 + 0.050i$
$R = \|r\|^2$	0.064	0.493	0.953	0.948
ϕ (rad)	3.016	0.000	0.000	0.000

Table 4.1: Reflection coefficients for the four input states of the atom-photon gate, simulated via a time-domain simulation outlined in Appendix A using the system parameters characterized in section 5.1. The atomic states $|0\rangle$ and $|1\rangle$ correspond to the uncoupled and coupled ground states, respectively.

Table 4.1 shows that the reflection amplitude for the V polarized photon, is much bigger than that of the π polarized photon, thus giving the following relationship: $|r_V| > |r_{\pi,\text{coupled}}| > |r_{\pi,\text{uncoupled}}|$.

This amplitude mismatch ultimately reduces the overlap with an ideal CNOT gate truth table. However, since the cavity parameters are fixed by the requirements of the broader *CavityX* experiment, the atom-state-dependent amplitude difference cannot be addressed independently, as this would require changing the cavity itself. The following sections therefore focus on compensating the polarization-dependent mismatch, which can be corrected externally.

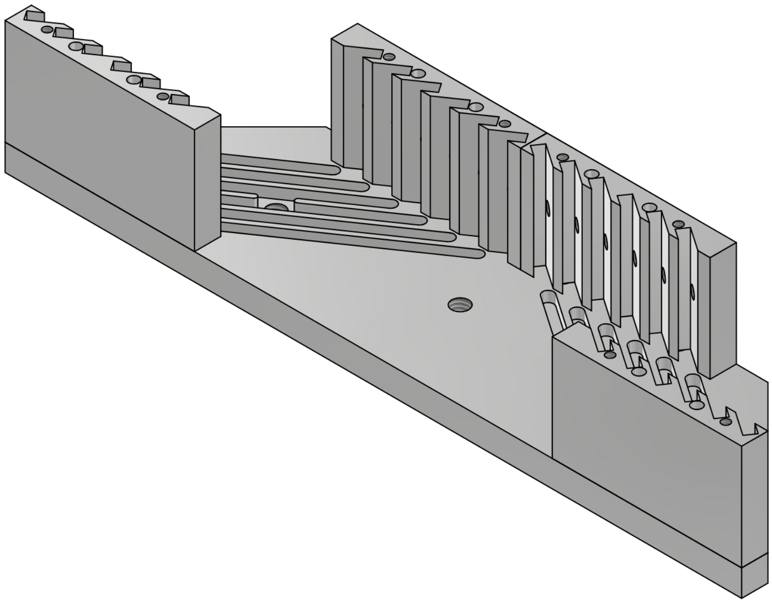
4.2 Inducing Losses via a Brewster Window Array

A Brewster window is a piece of uncoated UV fused silica. When placed at the so-called Brewster angle relative to the incoming light, then the p-polarized portion gets transmitted without any losses, while the s-polarized portion gets partially reflected. This effectively reduces the transmission through the optical element for s-polarized light. Here, the p-polarized light is identical to the π polarized light, whereas the s-polarized light matches that of the V polarized light.
As previously discussed in section 4.1, in order to maximize the overlap of the measured CNOT gate truth table with that of an ideal CNOT gate, additional polarization-dependent losses must be included in the V mode, to match those the π mode experiences.
In order to achieve this an array of Brewster windows were installed on a custom machined mount, which place the Brewster windows at the Brewster angle relative to the incoming light, so that the V polarized portion of the incoming light gets attenuated prior to interacting with the atom-cavity system. This ensures that in reflection, both polarization modes have the same reflection amplitude, allowing for the observation of the flip on the photonic qubit in the CNOT basis.
The Brewster angle can be simply calculated via:[47]

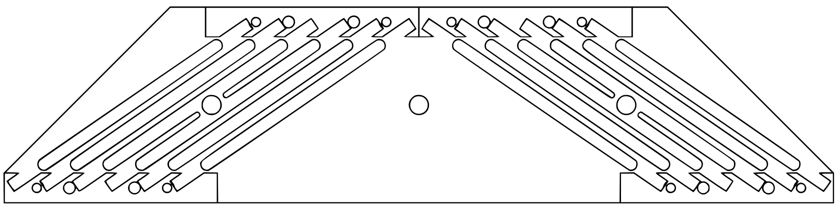
$$\tan\theta_B = n_t/n_i \tag{4.1}$$

$$n_t^2 - 1 = \frac{0.6961663\lambda^2}{\lambda^2 - 0.0684043^2} + \frac{0.4079426\lambda^2}{\lambda^2 - 0.1162414^2} + \frac{0.8974794\lambda^2}{\lambda^2 - 9.896161^2} \tag{4.2}$$

In (4.2) n_t is the refractive index of the Brewster window, and n_i would be the refractive index from which the incoming light is coming from (here: air $n_i \sim 1.004$). For light that is resonant to the 2-1' transition of the D_2-line of ^{87}Rb, $n_t = 1.453667$, resulting in a Brewster angle $\theta_B = 55.467288°$. Measurements outlined in section 5.1.2 determined that the average transmission through a single Brewster window placed at the Brewster angle θ_B to be $99.98 \pm 0.17\%$ and $77.40 \pm 0.13\%$ for the H and V polarization respectively.
The custom Brewster window mount as mentioned before has the main functionality of placing the elements in an array at the Brewster angle θ_b relative to the incoming light. It can house up to 12 Brewster windows when using the screw hole in the middle or 8 Brewster windows when using the screw holes on each side of the mount. The latter allows the mount to be placed on two separate posts, making the overall construction more mechanically stable, which eventually ended up being the configuration used in this project. This configuration not only provided greater mechanical stability, but was also sufficient for the experiment, since the optimal CNOT overlap was found to lie between 6 and 7 Brewster windows, as determined by a time-domain simulation of the atom-cavity system (see Appendix A.2).
Furthermore, the Brewster window mount has two sections in which the layout of one half is mirrored to that of the other half. The reason being that each Brewster window laterally displaces the transmitted beam by a fixed amount due to refraction through the tilted plate. This significantly reduces the coupling into the cavity fiber. In order to combat this, the two halves of the mount are symmetrical to one another, so that the shift of the optical beam path on both sides cancel each other out. By reversing the tilt direction in the second half of the array, each displacement is compensated by the corresponding window in the mirrored section. Although this is only the case when the total amount of Brewster windows is even and both halves are balanced, at most the shift would be equal to that of a single Brewster window shifting



(a) Isometric view.



(b) Top-down view.

Figure 4.1: Custom-machined mount for Brewster windows. The mount features two mirrored sections of angled slots, each holding up to six Brewster windows at $\sim 55.48°$.

the optical beam path. This can be compensated by realigning the input beam into the cavity fiber.

4.3 Photon Qubit Input-Output Path

For the atom-photon gate, weak coherent pulses are used in order to simulate the effect of single photons getting reflected from the cavity. It is therefore imperative to precisely control the power going into the cavity, and also the frequency of the light. For that specific task the following setup was constructed.

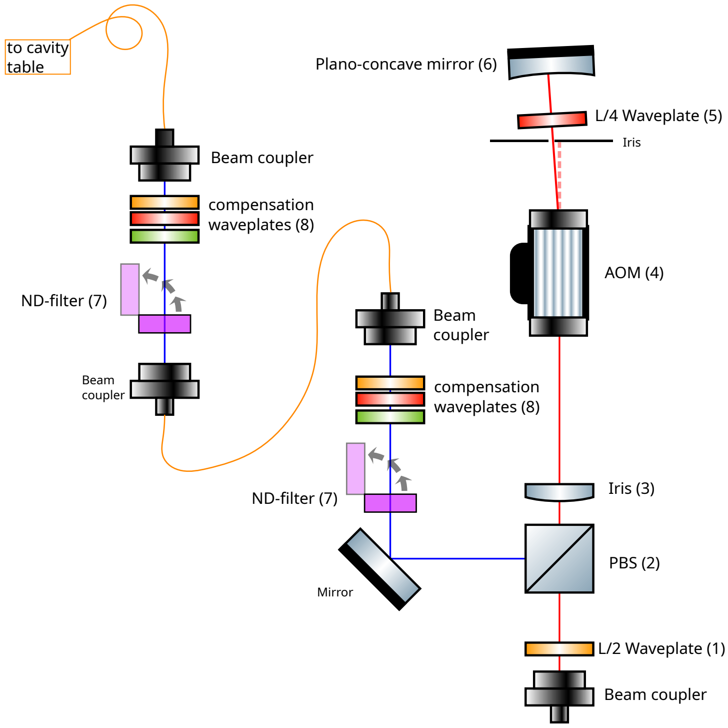


Figure 4.2: Double-pass AOM setup on the AOM table.

Figure 4.2 showcases a double-pass AOM track. The light going into the AOM track, is blue detuned to the 2-1' transition of the D_2-line by 480 MHz. Part of the light (not shown here), can then be superimposed either with a stable reference frequency source like an optical frequency comb from the company *MenloSystems GmbH* when the laser frequency does not need to be scanned, or with another laser, for when it does need to be scanned, e.g., for performing a normal-mode spectroscopy. The polarization of the light first gets set by a $\lambda/2$ retardation waveplate (1) in order to pass through the polarizing beam splitter (2) in full transmission. A plano convex lens (3) focuses the light into an acousto-optic modulator (4). In the double-pass configuration shown here, the AOM shifts the optical frequency by twice its driving frequency, enabling continuous frequency tuning.

The zeroth order of the transmitted light will get blocked by an iris, whereas the minus first order of the light is able to pass through. In order to now fully get reflected into the other arm of the PBS (marked in blue), the light passes through a $\lambda/4$ retardation plate (5) twice, while getting reflected and focused back into the AOM via a plano-concave mirror (6).
The light then passes through two sets of neutral density (ND) filters (7) (each set transmitting only $\simeq 0.037\%$) and polarizers plus one $\lambda/2$ and $\lambda/4$ retardation waveplates (8) which allow for the alignment of the light with the principal axis of the polarization-maintaining (PM) fiber. The two sets of ND filters, can be selectively inserted into or removed from the beam path, attenuating the pulses to mean photon numbers in the range $\bar{n} = \{1\mathrm{e}{-5}, \ldots, 4\}$ at the cavity input, with both inserted. The light then gets sent to the atomic-table via another PM fiber.
Once the light goes from the AOM table to the atom table, then the setup shown in Figure 4.3 is

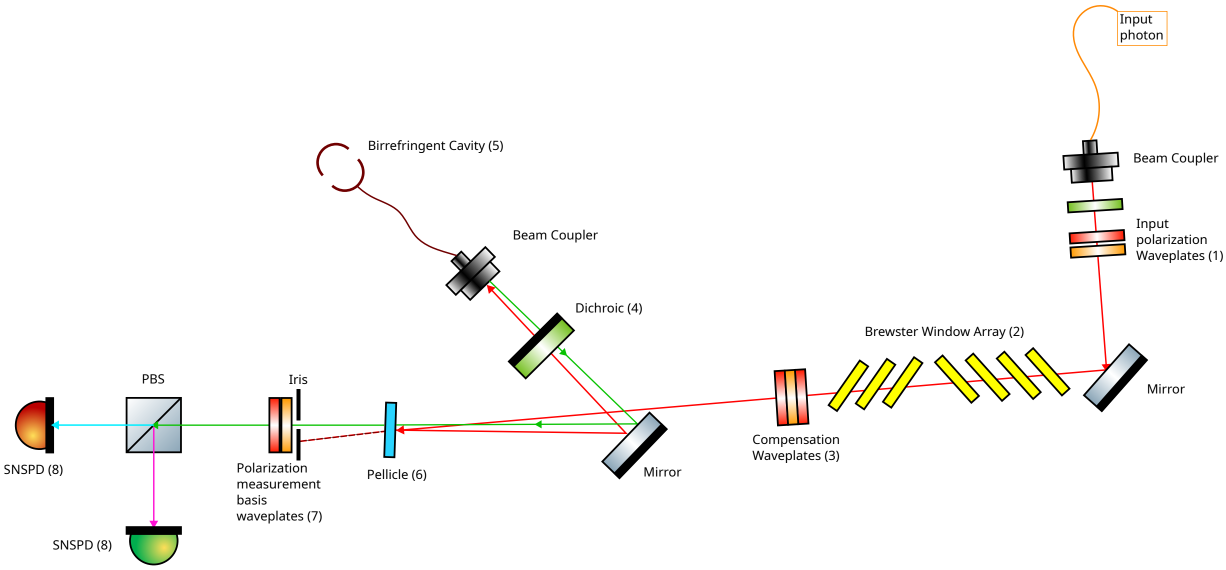


Figure 4.3: Optical setup on the atom-cavity table. The input photon is highlighted in red, and the reflected photon is highlighted in green.

used. The input polarization for the atom-photon gate is able to be set via two automatically controlled retardation waveplates (1), with the light coming from the AOM table. In order to compensate for the polarization-dependent losses the light then passes through the array of Brewster windows (2). A set of compensation waveplates (3) then ensure that the H and V polarized light interacting with the atom-cavity system, are mapped to the π and V eigenmodes of the cavity. In order to couple the light into the cavity fiber, it is reflected off of a pellicle beam splitter (6). The pellicle beam splitter reflects 1% of the beam onto a mirror and through a dichroic mirror (4) into the cavity fiber coupler. The dichroic mirror also serves to couple the intracavity dipole trap light (not shown) into the same fiber. The light reflected from the birefringent cavity (5), passes through the dichroic mirror and reflects off the mirror back onto the pellicle. Since the pellicle transmits $\sim 99\%$ of the returning light, the reflected signal continues with minimal loss (green path) to the detection stage. An arbitrary polarization measurement basis can then be set via two automatically controlled retardation waveplates (7). Finally the light passes through another PBS, and the two arms couple into two separate superconducting nanowire single-photon detectors (SNSPDs,8). The efficiency of the entire detection setup has an estimated 74.21% at an average dark count rate of 43 Hz per SNSPD channel.

Prior to building the setup shown in Figure 4.3, the gate photons were routed via an existing beam path known as the probe beam, which had originally been used for empty-cavity spectroscopy to determine the mode-matching efficiency and photon decay rate.
The earlier setup can be seen in Figure 4.4. The light coming from the probe beam would be coupled into the cavity-fiber, through the transmission and subsequent reflection dichroic mirrors (4). Each of these mirrors is a *LL01-785-12.5 MaxLine laser clean-up filter* from the company *Semrock*, and possess the feature of transmitting most of the incoming light and reflecting only a small portion. This setup was chosen so that both the intracavity light trap (10) with its own set of compensation waveplates (7) and the probe beam, could be coupled into the fiber simultaneously.

The problem of this initial setup was, that precise control over the polarization and its losses is crucial, especially after reflection from the atom-cavity system. However, as it was the case with this setup, the coupling into the cavity fiber, was achieved via reflection off a dichroic mirror, which introduced additional polarization-dependent losses and rotations in the s and p polarization. In order to account for extra rotations induced by the dichroic mirrors (4), another set of compensation waveplates was included right in front of the cavity fiber coupler (5). This then meant that the first set of compensation waveplates (3), was meant to account for the rotations induced by the two dichroic mirrors, thus mapping the H and V polarization of the incoming light to the s and p polarization of the two mirrors. The second set of compensation waveplates (5)

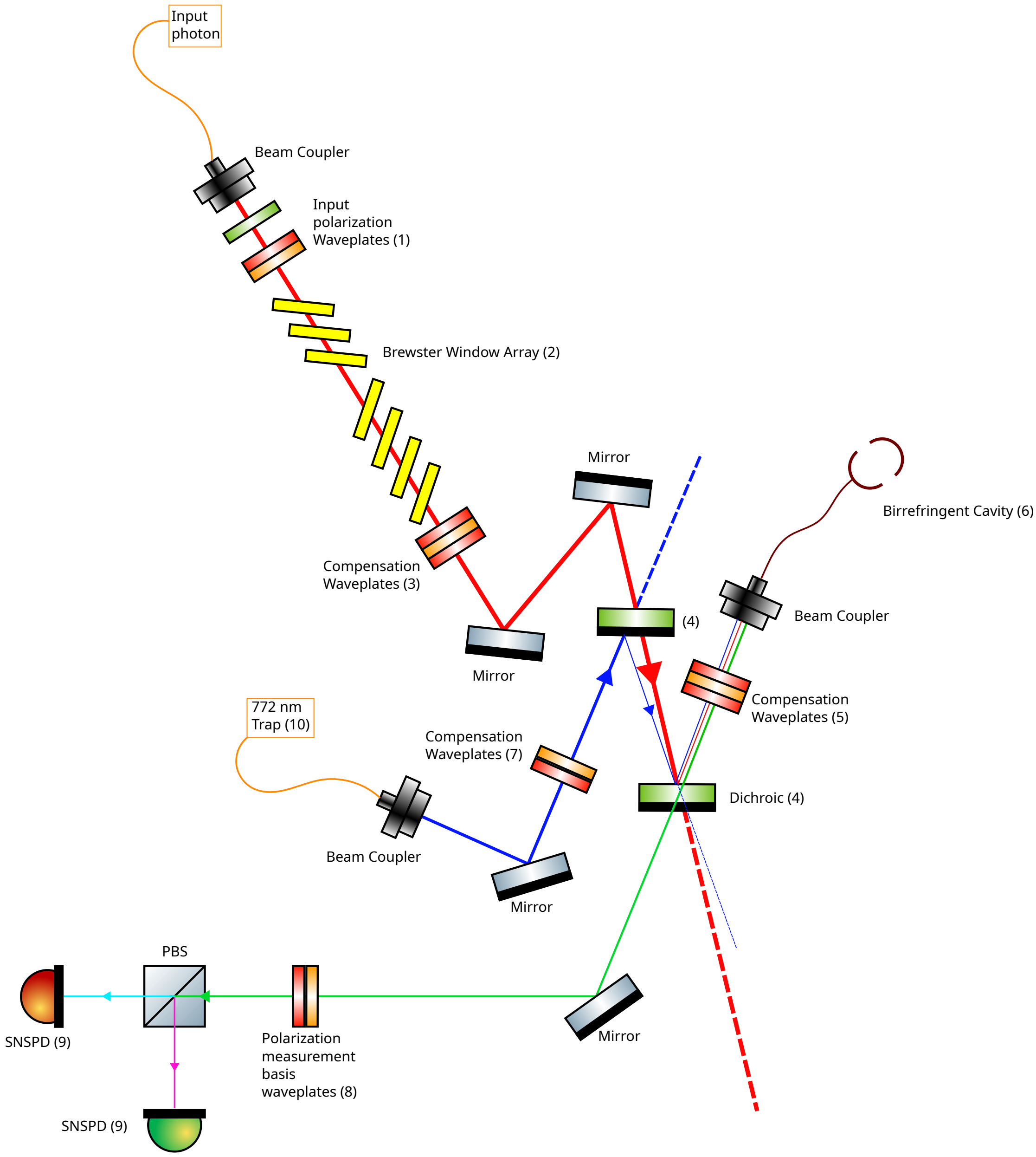


Figure 4.4: Old optical setup on the atom-cavity table. The input photon is highlighted in red, the reflected photon is highlighted in green, and the 772 nm short cavity trap light is highlighted in blue.

then mapped the s and p polarizations of the dichroic mirrors to the π and V mode of the cavity. However, with no means to check the polarization of the reflected light from the dichroic mirror via a polarimeter, due to spatial constraints, this setup proved to be unreliable. The measurement setup with the automatic waveplates (8) and two SNSPDs (9) would have remained the same.

Finally, by changing to the setup shown in Figure 4.3, and instead opting to use a pellicle, which had much smaller polarization-dependent losses and rotations in reflection compared to the dichroic mirrors, this issue was resolved. Furthermore, it was verified that the polarization-dependent changes both in reflection and transmission through the pellicle were minimal, thus ensuring that the correctly reflected light will not receive further penalties when going to the detectors.
A quantitative comparison of the two configurations highlights the reduction of polarization-dependent losses when replacing the dichroic mirrors with a pellicle beam splitter. In the original setup, a pronounced imbalance between H and V polarizations is observed before the coupler, with relative transmission differences on the order of ~ 20–30%, indicating significant polarization-dependent loss introduced by the dichroic elements. In contrast, for the pellicle-based configuration, the normalized transmissions for H and V are closely matched, corresponding to only a few fractions of a percent difference. This demonstrates that the pellicle substantially suppresses polarization-dependent losses compared to the old dichroic mirror configuration.

Chapter 5

Atom-Photon Quantum Logic Gate

Using the setup explained in chapter 4, one can characterize important experimental parameters in order to estimate the cavity coupling as well as its losses when reflecting light off the atom-cavity system. Furthermore, using these values, the overlap with an ideal CPF/CNOT gate truth table can then be estimated and compared to the measured values.
Section 5.1 of this chapter goes over the experimental parameters which are relevant for the gate mechanism and how they were determined. Section 5.2 reports on the performance of the atom-photon gate in both the CPF and CNOT bases, alongside a comparison between the experimental and simulated results for the Brewster window and mean photon number scans. Section 5.3 then demonstrates the coherent behavior of the gate through the generation of a Bell state. Finally, an analysis of the limitations currently imposed by the system is given in Section 5.4.

5.1 Experimental Parameters relevant for the gate mechanism

In order to estimate the overlap between the experimentally realizable and ideal CNOT gate, one first needs to determine various experimental parameters. Using the experimental parameters one can then calculate the reflectivity for each of the input states of the atom-photon gate, and use those to calculate a theoretical overlap with an ideal CPF/CNOT gate.

5.1.1 Cavity coupling and losses

The first set of experimental parameters to extract are the total cavity decay rate κ, the mode-matching parameters μ_{fr} and μ_{fc}, and the atom-cavity coupling strength g. These are obtained from two complementary spectroscopy measurements on the locked cavity: an empty-cavity spectroscopy (without a trapped atom) to determine κ, μ_{fr}, and μ_{fc}, and a normal-mode spectroscopy (with a coupled atom) to extract g.
For the empty-cavity measurement, the probe laser is scanned across the π-mode resonance of the cavity while being phase-modulated at 300 MHz by an electro-optic modulator (EOM). The EOM sidebands appear as additional peaks in the transmission spectrum, separated by a known frequency from the carrier, and thereby provide an absolute time-to-frequency calibration of the scan axis. The π-mode of the cavity is locked during the measurement to the $F = 2, m_F = 0 \leftrightarrow F' = 1, m_F = 0$ transition, and both the transmitted and reflected signals are recorded simultaneously. Since no atom is loaded, the coupling term in equation (2.10) vanishes ($g = 0$). The transmission spectrum is fitted with a sum of Lorentzian lineshapes. Each resonance peak is described by a cavity transfer function of the form:

$$T(\omega) = \left| \frac{A}{i(\omega - \omega_c) + \kappa} \right|^2, \tag{5.1}$$

where A is a fit amplitude, ω_c the resonance frequency, and κ the half-width at half maximum (HWHM) of the cavity mode. The full linewidth 2κ is extracted from the fit and averaged over multiple successive cavity scans, yielding $2\kappa = 2\pi \times (116.19 \pm 0.37)$ MHz.

The reflection spectrum is fitted using the empty-cavity limit of Eq. (2.10). Setting $g = 0$ and absorbing a relative phase into the mode-matching coefficient μ_{fc}, the normalized reflection amplitude simplifies to:

$$r(\omega) = \mu_{fr} - \mu_{fc}^2 \frac{2\kappa_{\mathrm{oc}}}{i(\omega - \omega_c) + \kappa}, \tag{5.2}$$

The fit yields the fiber-cavity mode-matching parameter $\mu_{fc} = 0.873 \pm 0.002$ and the free-space mode-matching parameter $\mu_{fr} = 0.978 \pm 0.006$. The outcoupling ratio $\kappa_{\mathrm{oc}}/\kappa = 0.85$ is held fixed during the fit, as this value has been consistently measured across multiple cavity assemblies over several years. The resulting fit parameters are summarized in Table 3.1 in section 3.1, and representative transmission and reflection spectra with their respective fits are shown in Figure 5.1.

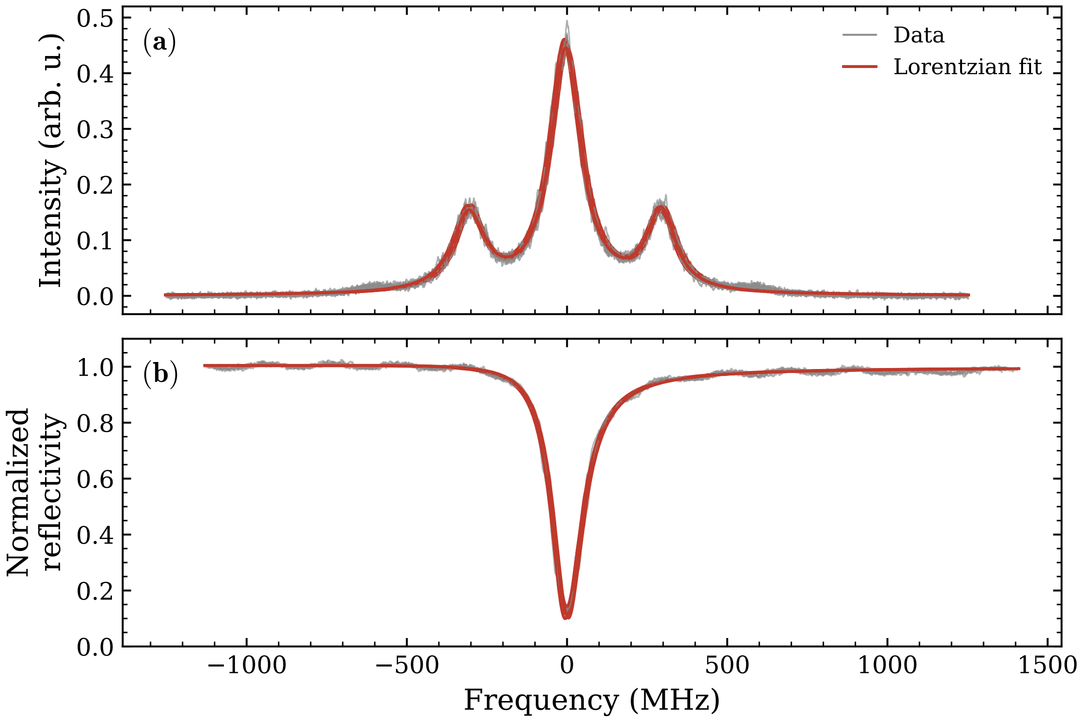


Figure 5.1: Empty-cavity spectroscopy of the π-mode. The probe laser frequency is scanned across the cavity resonance while phase-modulated at 300 MHz by an electro-optic modulator, producing sidebands that serve as frequency markers for the time-to-frequency calibration. **(a)** Cavity transmission spectrum. Multiple successive scans (grey) are individually fitted with Lorentzian lineshapes (red) according to Eq. (5.1), from which the full linewidth 2κ is extracted. **(b)** Normalized cavity reflection spectrum. The reflection dips are fitted using Eq. (5.2) to extract the mode-matching parameter μ_{fc}. Each scan is centered on its respective resonance frequency.

With κ and the mode-matching parameters determined, the remaining unknown is the atom-cavity coupling strength g. This is extracted from a normal-mode spectroscopy measurement, in which the probe laser is scanned across the cavity resonance while a single atom is trapped and coupled to the π mode.

After trapping the atom it is cooled and prepared in the $F = 1, m_F = 0$ Zeeman state via the aforementioned methods in section 3.3. One can then either record the normal-mode spectroscopy for the uncoupled ($F = 1, m_F = 0$) or coupled state ($F = 2, m_F = 0$), by optionally applying a microwave π-pulse that transfers the population from the uncoupled to the coupled state before the probe scan. The probe laser, which is resonant to the $F = 2, m_F = 0 \leftrightarrow F' = 1, m_F = 0$ transition, is deployed after the preparation of the atom. For each point of the normal-mode spectroscopy, the probe laser is deployed for $1\,\mu$s.

For the normal-mode spectroscopy the probe laser has to be scanned over a large frequency range in order to fully capture the entire reflection spectrum. This is achieved by changing the locking point (the reference detuning), which then allows its frequency to be changed in a controlled manner. The beat signal to which the laser will then be locked is supplied by an arbitrary waveform generator (AWG), which continuously changes the frequency over a range of 25 MHz to 300 MHz. Here, locking the laser to a beat frequency of 25 MHz would make the probe laser red-detuned to the $2 - 1'$ transition, whereas a beat frequency of 300 MHz would make it blue-detuned to the respective transition.

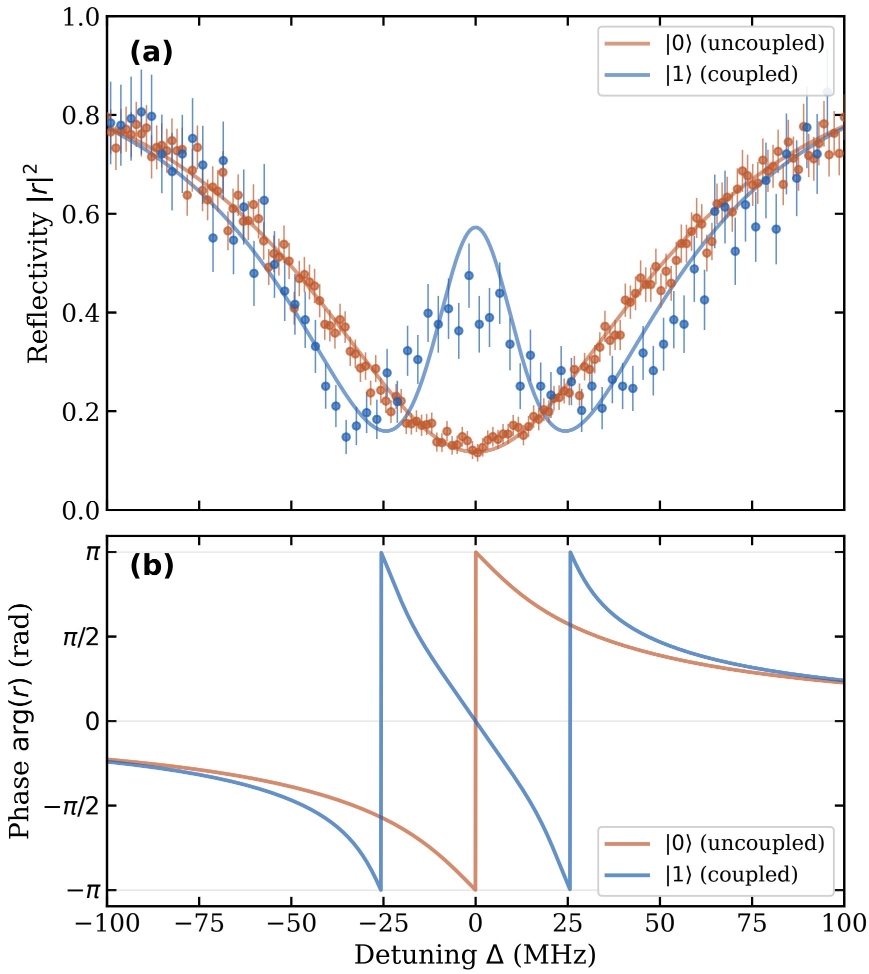


Figure 5.2: Normal-mode spectroscopy of the atom-cavity system. **(a)** Normalized reflectivity $|r|^2$ as a function of probe detuning Δ for the uncoupled state $|0\rangle$ (red) and the coupled state $|1\rangle$ (blue). The uncoupled spectrum shows the bare cavity resonance, fitted with the empty-cavity reflection coefficient, as the uncoupled state is off-resonant from the probe, so the system behaves as an empty cavity. The coupled spectrum exhibits the vacuum Rabi splitting, fitted using equation (2.10), yielding a coupling strength of $g = 2\pi \times (25.9 \pm 0.5)$ MHz. **(b)** Phase $\arg(r)$ of the reflected field simulated from the respective fit models. The case for the atom in the uncoupled state showcases a phase shift of π on resonance, while the coupled system features a phase shift of 0.

From the fit in Figure 5.2 an atom-cavity coupling $g = 2\pi \times (25.9 \pm 0.5)$MHz can be extracted. With these parameters in hand, the next step is determining the optimal V-mode attenuation.

5.1.2 Brewster plate attenuation

Using the values determined in section 5.1.1, one can simulate the reflection amplitudes for each of the input states of the gate using the time-domain simulation described in Appendix A. Using the simulation results, one can simulate the ideal attenuation of the almost ideally reflected V polarized portion of the input light, in order to achieve the highest possible overlap with the ideal CNOT truth table (Appendix A.2). From the simulated overlap, one deduces that the optimal

transmission of the V polarized light is at $T_V \approx 24\%$. Using this information, the ideal number of Brewster windows can thus be determined.
For the characterization of the Brewster windows themselves, two measurements were conducted. First, only a single Brewster plate was inserted into the custom mount to check the average attenuation across a set of Brewster windows in the V polarization. This ensures that all Brewster windows operate up to spec, and that the attenuation of the V polarization is happening in constant jumps. At the Brewster angle θ_B the transmission through a single Brewster window was $99.98 \pm 0.17\%$ and $77.40 \pm 0.13\%$ for the H and V polarization respectively.
The second characterization that needed to be performed places multiple Brewster windows in an array to closely resemble the setup used in the actual experiment. In this configuration, one can then measure the powers in transmission both for H and V to determine the ideal amounts of Brewster windows one would need to use in order to achieve the highest gate fidelity.

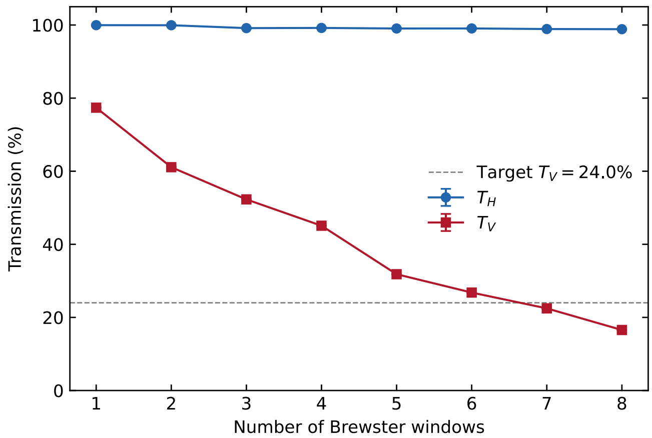


Figure 5.3: Measured transmission of H- and V-polarized light through N Brewster windows placed in series at Brewster's angle. The H-polarization transmits near-unity for all configurations, while the V-polarization is progressively attenuated. The dashed line indicates the target V-mode transmission of 24.0 % for optimal overlap with the ideal CNOT truth table.

In Figure 5.3, one can see that the configuration for the optimal overlap with the ideal CNOT truth table lies between 6–7 Brewster windows.

5.2 Atom-Photon gate performance

This section will now go over the measurements for the gate truth tables. First the experimental protocol for the measurement as well as the resulting truth tables will be showcased in section 5.2.1. Afterward, the overlap of the measured truth table will be characterized as a function of the number of Brewster windows installed in the custom mount and the mean input photon number in section 5.2.2.

5.2.1 Gate Truth Tables

The measurement of the gate truth table starts with the preparation of the atom. Using the protocol outlined in section 3.3, one can choose to then prepare the atom in the coupled state $|F = 2, m_F = 0\rangle = |1\rangle$ by deploying a microwave π-pulse, or leave the atom in the uncoupled $|F = 1, m_F = 0\rangle = |0\rangle$ state. Once the atom is prepared in one of the atomic eigenstates, a gated weak coherent laser pulse will be sent to the cavity, which for the following measurements will have an input mean photon number of $\bar{n} \approx 0.1$. As explained in chapter 2, depending on the choice of the photonic qubit eigenbasis, one chooses to either realize a CPF gate (basis: $|\pi\rangle$ and $|V\rangle$) or a CNOT gate (basis: $|\pm\rangle = \frac{|V\rangle \pm |\pi\rangle}{\sqrt{2}}$). The CPF basis is implemented using the H and V polarizations, which are mapped to the π and V modes of the birefringent cavity via the first set of compensation waveplates shown in Figure 4.3. For the CNOT basis, one can then just use a set of orthogonal polarizations that are a superposition of H and V. In this case the CNOT basis was chosen to be the R and L basis. Once the photon pulse has been reflected off of the cavity, a state detection (SD) beam is deployed which produces SD-counts if the atom is in the coupled state $|1\rangle$, or none if the atom is in the uncoupled state $|0\rangle$. A detailed schematic of the protocol is shown in Figure 5.4.

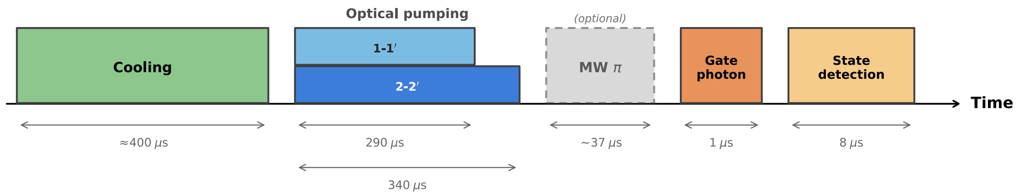


Figure 5.4: Experimental sequence for a single trial of the gate truth table measurement. Each trial begins with $\approx$400 μs of cooling, followed by optical pumping using two lasers addressing the $1 \to 1'$ and $2 \to 2'$ transitions simultaneously. The $1 \to 1'$ laser is switched off after 290 μs, with the $2 \to 2'$ laser remaining on for an additional 50 μs, preparing the atom in the $|F = 1, m_F = 0\rangle$ state. To prepare the atom in the coupled state $|1\rangle$, an optional microwave π pulse transfers population from $|0\rangle$ to $|1\rangle$ (dashed box). A weak coherent pulse of 1 μs duration and mean input photon number $\bar{n} \approx 0.1$ then probes the atom–cavity system, implementing the gate interaction. Finally, fluorescence-based state detection over 8 μs determines the atomic output state. Time axis is not to scale.

By then setting the detection waveplates from Figure 4.3 to measure the same basis as the input, and post-selecting on the cases where only a single photon was reflected during the photon-gate pulse, one is able to measure the truth tables for the CPF and CNOT gates shown in Figure 5.5. The truth tables for the CPF and CNOT gates show an overlap of $(92.41 \pm 0.52)\,\%$ and $(88.15 \pm 0.86)\,\%$ with the ideal truth tables, respectively. The overlap is defined as

$$\mathcal{O} = \frac{1}{4}\mathrm{Tr}\big(T_{\mathrm{ideal}}^{\top}\,T_{\mathrm{meas}}\big)\,, \tag{5.3}$$

where T_{ideal} and T_{meas} are the 4×4 ideal and measured truth table matrices, respectively. The errors are calculated by error propagation under the assumption of independent errors of different bars.

5.2.2 Mean photon number and Brewster window scan

To map out the parameter space of the gate and benchmark the underlying simulation model, the overlap with the ideal CNOT truth table is measured as a function of two key control parameters: the V-mode attenuation set by the Brewster-window array, and the mean input photon number $\bar{n}$. Both of these scans were simulated using QuTiP [48] with the exact details of the simulation outlined in Appendix A. The simulated scans are then compared to the experimentally determined overlaps, in order to showcase the limitations of the simulation itself and how well it reflects the actual experiment. A comparison for each respective scan and its simulation can be seen in Figure 5.6.
Comparing the experimental data to the simulated overlaps, the Brewster window scan follows the simulated trend; deviations from the curve are consistent with atom-cavity imperfections discussed in §5.4. As for the photon number scan, the simulated and measured overlaps agree within error up until a mean photon number of $\bar{n} \approx 1$, where the experimental data shows a much quicker drop in overlap when compared to the simulation. This indicates that while the simulation reproduces the experiment well for low mean input photon numbers, it underestimates the degradation at high photon numbers, where the experimental data shows higher atomic scattering (see section 5.4.5).

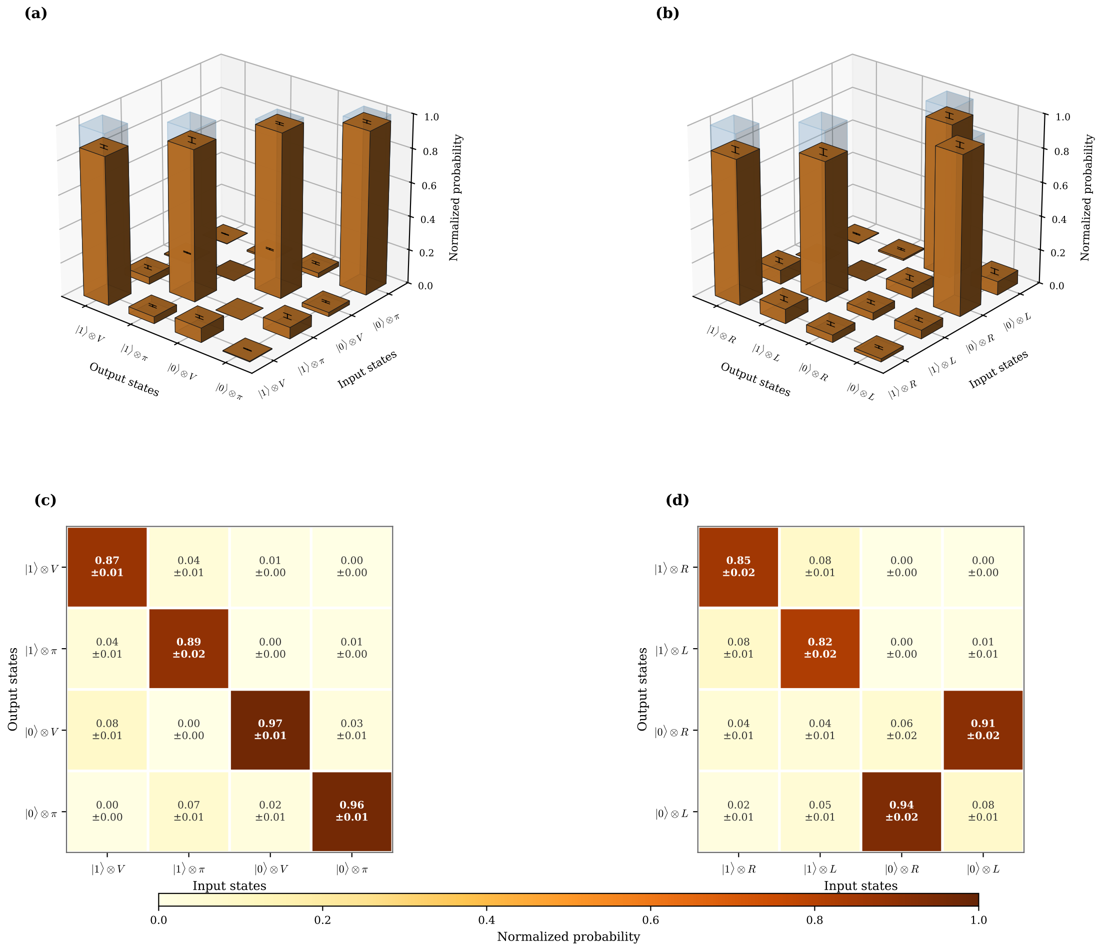


Figure 5.5: Measured truth tables of the atom-photon quantum gate with a mean input photon number of $\bar{n} \approx 0.1$. (a), (b) Three-dimensional bar charts showing the normalized output-state probabilities for all four input-state combinations. Semi-transparent blue bars indicate the ideal gate operation. (c), (d) Corresponding matrix representations with numerical values and statistical uncertainties. The left column shows the controlled-phase (CPF) gate in the $|\pi\rangle\,/\,|V\rangle$ eigenbasis, where the ideal operation is the identity on the computational basis states. The right column shows the controlled-NOT (CNOT) gate in the $|R\rangle\,/\,|L\rangle$ eigenbasis, where the photon polarization is flipped when the atom is in the uncoupled state $|0\rangle$. The atomic output state is determined via fluorescence-based state detection following the protocol shown in Figure 5.4.

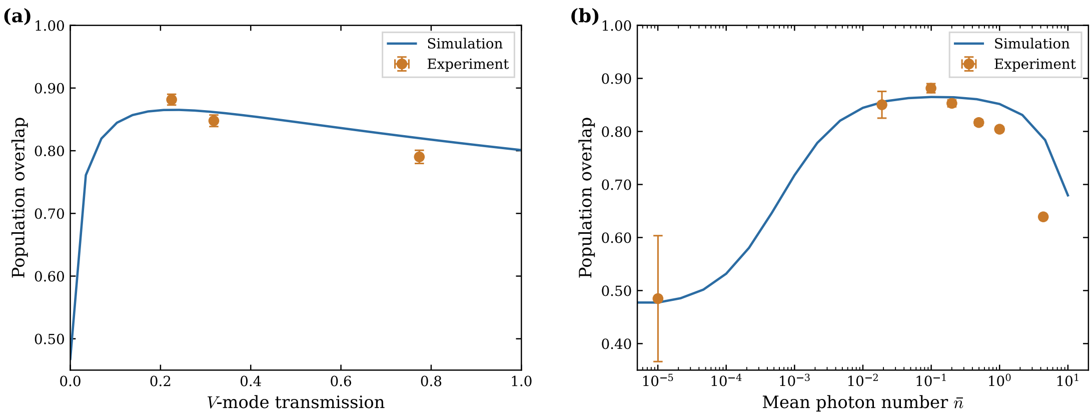


Figure 5.6: Comparison of the measured population overlap with the QuTiP master-equation simulation. (a) Population overlap as a function of the V-mode transmission, controlled by the number of intracavity Brewster windows. The three experimental points correspond to seven, five, and one Brewster windows (from left to right), with the mean input photon number fixed at $\bar{n} \approx 0.1$. (b) Population overlap as a function of the mean input photon number $\bar{n}$, measured with seven Brewster windows installed. The overlap peaks near $\bar{n} \approx 0.1$ and decreases for both lower photon numbers, where vacuum contributions dominate, and higher photon numbers, where multi-photon components degrade the gate fidelity. Solid lines show the simulation results; markers indicate experimental data.

5.3 Generation of Bell States

What distinguishes a quantum logic gate from its classical counterpart is its ability to generate entangled states from separable inputs. To demonstrate this behavior, the CNOT gate is applied to a separable input state in which the atom is prepared in a superposition, with the expected output being a Bell state. Concretely, the atom is prepared in $|+_a\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ via a resonant microwave $\pi/2$ pulse, after which a weak coherent pulse ($\bar{n} \approx 0.1$) of $|L_p\rangle$-polarized light is reflected off the atom-cavity system. The gate action then ideally produces the Bell state $|\Phi^+\rangle$:

$$|+_a, L_p\rangle \to |\Phi^+\rangle = \frac{1}{\sqrt{2}} \left(|0_a, R_p\rangle + |1_a, L_p\rangle\right). \tag{5.4}$$

The measurement is performed with seven Brewster windows installed, matching the optimal configuration identified in section 5.2.2.

The joint atom-photon state is characterized via quantum-state tomography: by measuring in three measurement bases for the photon and atom separately and applying maximum-likelihood estimation [49, 50], the full density matrix and the Bell state fidelity are extracted. Figure 5.7 shows the extracted density matrix.

The reconstructed density matrix reproduces the dominant features of the ideal $|\Phi^+\rangle$ state: population is concentrated on the diagonal elements $|0_a, R_p\rangle$ and $|1_a, L_p\rangle$, with coherences of comparable magnitude connecting them. However, both the populations and coherences are reduced from their ideal value of 0.5, and residual population has leaked into the wrong-parity states $|0_a, L_p\rangle$ and $|1_a, R_p\rangle$.

The fidelity with the expected state is $F = \langle\Phi^+|\,\hat{\rho}\,|\Phi^+\rangle = 65.90 \pm 1.73\%$, where the standard error has been determined with the Monte Carlo technique [49]. Although this does not violate the CHSH inequality, it places the fidelity above the 50% threshold confirming that entanglement was generated. This measurement represents a first demonstration of entanglement generation on the platform; a more systematic optimization of the experimental parameters and an extended integration time are expected to push the fidelity considerably higher.

5.4 Limitations and imperfections

The measured gate overlap and Bell state fidelity fall short of unity due to several independent imperfections, each contributing at a different scale. This section identifies and discusses the

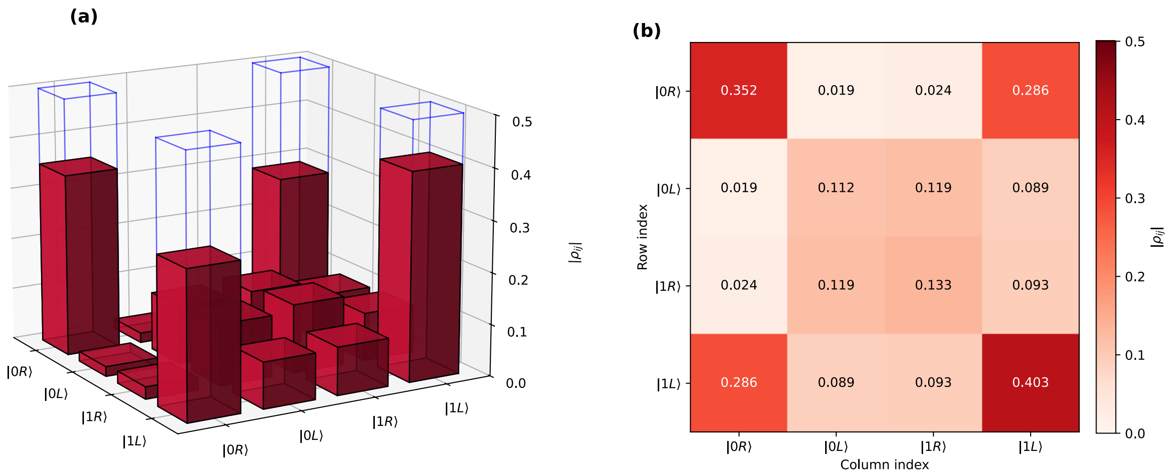


Figure 5.7: Reconstructed density matrix of the generated Bell state $|\Phi^+\rangle$. (a) Three-dimensional city plot of $|\rho_{ij}|$, with semi-transparent blue wireframe bars indicating the ideal density matrix. (b) Corresponding heatmap with numerical values for each matrix element.

dominant sources, ordered roughly by their impact on the gate performance.

5.4.1 Asymmetric reflection amplitudes

An ideal controlled-phase gate requires that the cavity reflects photons with equal amplitude for both atomic states, differing only in the acquired phase. As shown in Table 4.1, this condition is not met for the π-polarized mode: the uncoupled state $|0\rangle$ yields a reflectivity of $R_{0,\pi} = 0.064$, while the coupled state $|1\rangle$ yields $R_{1,\pi} = 0.493$. The large asymmetry arises because the system does not operate deep in the strong-coupling, single-sided-cavity regime where the empty-cavity reflection approaches unity. Instead, the cavity parameters place the uncoupled reflectivity well below the coupled one, so that even before any V-mode attenuation is applied, the two π-mode amplitudes are far from matched.

This mismatch directly limits the maximal overlap with the ideal CNOT truth table. Mitigating it would require a fundamentally different set of cavity parameters — for example, a higher mirror reflectivity on the incoupling side to push the empty-cavity reflectivity closer to unity — which would necessitate replacing the cavity altogether. Alternative routes include using a different atomic transition with a larger coupling strength on the available mode, or repositioning the fiber tips via the piezo-controlled mounts to modify the atom-cavity coupling g. None of these changes are trivial, and the current cavity parameters therefore set a hard ceiling on the achievable gate overlap.

A related but far less significant limitation is the discrete nature of the V-mode attenuation. The overlap is maximized at a specific V-mode attenuation that matches the π-mode reflectivity, but the Brewster windows used in the setup change the attenuation in fixed steps set by the window material, placing the current operating point roughly 2 % away from the theoretical optimum. As visible in Figure 5.6(a), the overlap curve is relatively flat near its maximum, so this offset has a negligible effect compared to the other imperfections discussed below. Nonetheless, for a future implementation targeting near-unity overlap, a continuously tunable V-mode attenuator would eliminate this residual limitation entirely.

5.4.2 Imperfect atomic state preparation

The gate protocol assumes that the atom is initialized in the coupled state $|F{=}2, m_F{=}0\rangle = |1\rangle$ with unit probability. Experimentally, the preparation proceeds in two steps: Zeeman optical pumping into $|F{=}1, m_F{=}0\rangle = |0\rangle$, followed by a resonant microwave π pulse that transfers the population to $|F{=}2, m_F{=}0\rangle$. Neither step is perfect. The combined preparation probability in

the present dataset is approximately 88 %, meaning that roughly 12 % of trials begin with the atom still in the uncoupled state $|0\rangle$.
Because an incorrectly prepared atom does not interact with the cavity mode in the intended way, these trials contribute incoherently to the measurement and directly reduce the observed overlap. The effect is substantial: even if every other aspect of the gate were ideal, a 12 % preparation error would already lead to an error of $\sim 6\%$. Improving the Zeeman pumping – for instance by optimizing the pumping light polarization, duration, or magnetic field alignment – and improving the microwave pulse fidelity would therefore yield a significant and relatively straightforward gain in gate performance.

5.4.3 Thermal sensitivity of the detection basis

Extracting the gate overlap requires measuring the reflected photon in a well-defined polarization basis, set by the wave plates and polarizing beam splitter before the detectors. In practice this basis is not perfectly stable: when the UV LED used for atom loading during the MOT sequence is active, the measured extinction ratio between the two detection channels slowly drifts away from its calibrated value, and only recovers once the lamp is switched off. This effect is directly visible in Figure 5.8.

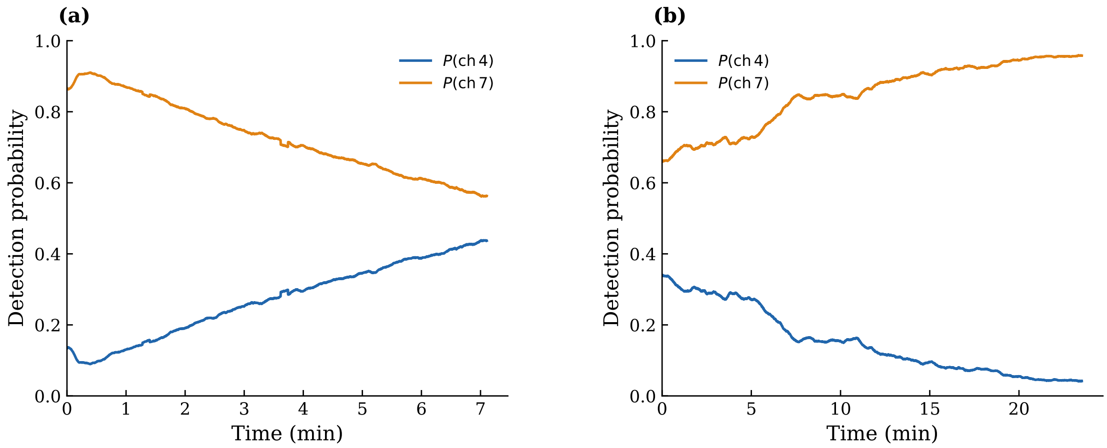


Figure 5.8: Long-term drift of the polarization extinction ratio measured on the gate photon reflection path, recorded at the two SNSPD channels of the polarization analysis stage (Ch 4 and Ch 7) when sending L polarized light. In each panel, the per-bin detection probabilities $P(\mathrm{ch}\,i) = N_i/(N_4 + N_7)$ are shown for Ch 4 (blue) and Ch 7 (orange), computed from count rates smoothed in 25 ms bins. The corresponding extinction ratio is $\eta = N_{\mathrm{Ch}\,7}/N_{\mathrm{Ch}\,4} = P(\mathrm{ch}\,7)/P(\mathrm{ch}\,4)$. **(a)** Drift recorded with the UV LED switched on: the imbalance between the two channels grows sharply within the first minute and then relaxes monotonically as $P(\mathrm{ch}\,4)$ rises and $P(\mathrm{ch}\,7)$ falls, indicating a continuous rotation of the analysed polarization. **(b)** Drift recorded after a 5 minute measurement where the atom was prepared in $|0\rangle$ and L polarized light was sent: $P(\mathrm{ch}\,4)$ decreases steadily from approximately 0.33 to 0.04 while $P(\mathrm{ch}\,7)$ increases correspondingly from 0.67 to 0.96, which corresponds to the extinction ratio rising from $\eta \approx 2$ to $\eta \approx 23$.

The origin of this drift is thermal, but not trivially localized. Three potential heat sources are at play in the setup: the rubidium dispenser, the UV LED, and the 852 nm vertical trap laser. The rubidium dispenser is the most disruptive of the three, inducing a large and easily resolvable shift of the cavity resonance when active [42]; for this reason it is only turned on briefly in the morning, and the gate measurements are started after the setup has been allowed to cool for several hours. Any residual dispenser-induced drift has therefore long decayed by the time a measurement run begins, and no separate characterization of dispenser heating is shown here. The vertical trap laser was additionally checked by recording the extinction ratio with only the trap active during a typical measurement window; no drift was observed above the measurement noise, and the corresponding trace is omitted for the same reason. This leaves the UV LED as the only heat source whose duty cycle changes on the timescale of a single measurement run, consistent with the drift in Figure 5.8 being UV-correlated. However, UV illumination does not measurably shift the cavity spectrum itself [42], so the drift cannot be attributed to a rotation of the birefringent eigenmodes of the cavity. The most likely remaining culprits are components in the immediate vicinity of the cavity that are not probed by the reflection spectrum – for instance stress-induced birefringence in the cavity fiber close to its tip, or thermally induced strain in the piezo-mounted fiber holder. Pinpointing the exact component is beyond the scope of this thesis. Regardless of its microscopic origin, the drift has a practical consequence for the gate measurements. Because the gate measurement window follows shortly after atom loading – and thus shortly after UV exposure – the residual thermal offset can introduce a systematic error in the measured overlap that varies from run to run depending on the thermal history. This is mitigated by reducing the MOT loading time down to 1 s, as this is where the trapping probability for a single atom is sufficiently high while keeping the time that the UV lamp is turned on at a minimum. However, even with these precautions in place, residual thermal drifts accumulated over the course of each measurement, requiring a cooldown period between runs (1.5 min measurement time vs. $\geq$5 min cooldown) before the extinction ratio recovered to its initial value.

5.4.4 Fluctuations in atom-cavity coupling

The atom is confined in an intracavity dipole trap, but its position is not perfectly fixed between successive trapping events. Small variations in the trapping position translate directly into variations in the atom-cavity coupling strength g, which depends on the local cavity mode amplitude at the atom's location. Since the reflection coefficients, the scattering rate, and consequently the gate overlap all depend on g, run-to-run position fluctuations introduce scatter around the value predicted by the simulation, which assumes a single fixed coupling strength.
This effect explains why some experimental data points in Figure 5.6 lie slightly above and others slightly below the simulated curve. A better-coupled atom yields a higher reflectivity contrast and therefore a higher overlap, while a worse-coupled atom yields the opposite. The fluctuations also affect the optical pumping efficiency, since the coupling to the pumping light likewise depends on the atom's position. As long as the distribution of trapping positions is centered near the assumed value, the experimental data are expected to scatter around the simulation without systematic bias, which is consistent with the observed trend.

5.4.5 Deviation at high mean photon numbers

At high mean photon numbers, the experimental data deviate from the simulation strongly, while at low photon numbers they all lie on the simulated curve (Figure 5.6(b)). While both the experiment and simulation show a degradation of the overlap with increasing photon number due to atom-induced scattering, the experimentally observed atomic scattering is higher than the simulation predicts, leading to a steeper drop in overlap than expected. The origin of this effect remains to be identified.

5.4.6 Additional limitations of the Bell state measurement

The Bell state fidelity of $(65.90 \pm 1.73)\,\%$ reported in Section 5.3 lies noticeably below the CNOT truth table overlap of $(88.15 \pm 0.86)\,\%$, despite both measurements being performed on the same system. Two factors contribute to this gap. First, the Bell state measurement is sensitive not only to the atomic populations – the only quantity probed by the truth table measurement – but also to the coherences between atom and photon, which are degraded by any dephasing process occurring between state preparation and readout. Second, the full tomography requires additional resonant microwave pulses to rotate the atomic state into the X and Y measurement bases, each of which carries a finite pulse fidelity and extends the total sequence duration. The longer measurement window also increases exposure to the thermal drifts discussed in Section 5.4.3. Although a more precise breakdown of the individual contributions cannot be given at present, the fidelity is expected to improve significantly with further optimization of the experimental parameters.

Chapter 6

Summary and Outlook

This thesis presents the realization of an atom-photon quantum gate between a single ^{87}Rb atom trapped in a high-finesse birefringent Fabry-Pérot cavity and a polarization-encoded photonic qubit, realized experimentally as a weak coherent pulse. This represents a further step towards a fully functional quantum network node on the CavityX platform, complementing the previously shown heralded atom-photon entanglement [26] and quantum memory capabilities [27] of the setup.

The gate mechanism exploits the strong cavity birefringence, which splits the polarization eigenmodes by $\Delta_{\text{biref}} \approx 2\pi \times 505\,\text{MHz}$, and the atom-state-dependent reflection of the π-polarized mode. An incoming π polarized photon acquires a π phase shift when the atom is prepared in the uncoupled clock state $|0\rangle = |F{=}1, m_F{=}0\rangle$, but no phase shift when the atom is prepared in the coupled clock state $|1\rangle = |F{=}2, m_F{=}0\rangle$. Using this I have implemented a controlled-phase-flip gate with the photon being encoded in the $\{|H\rangle, |V\rangle\}$ basis, where $|H\rangle$ is the polarization that drives the π atomic transition and will henceforth be denoted $|\pi\rangle$. By further preparing the atom in the superposition basis $\{\pm \equiv \frac{|V\rangle \pm |\pi\rangle}{\sqrt{2}}\}$, I was able to realize a controlled NOT gate using the same working principles. Because the cavity's V-mode is off-resonant, a photon with such polarization is reflected with near-unity amplitude. On the other hand, the π-mode suffers intracavity losses as its reflection amplitude is determined by the atomic qubit state. This imbalance in reflection amplitudes between the two polarizations would lead to a loss in overlap with the ideal gate's truth table. Thus, a custom mount holding an array of UV fused silica Brewster windows was installed in the input path to attenuate the V-polarized component externally and match the two reflection amplitudes.

The cavity and atom-cavity parameters were characterized experimentally, yielding an atom-cavity coupling strength of $g = 2\pi \times (25.9 \pm 0.5)\,\text{MHz}$. A master-equation simulation was used to determine the optimal mean input photon number $\hat{n} \approx 0.1$ and V-mode transmission $T_V \approx 24\,\%$ (corresponding to six to seven Brewster windows) for gate operation.

Using weak coherent pulses with a mean input photon number of $\bar{n} \approx 0.1$ and a configuration with seven Brewster windows installed in the custom Brewster mount, overlaps of $(92.41 \pm 0.52)\,\%$ with the ideal CPF truth table and $(88.15 \pm 0.86)\,\%$ with the ideal CNOT truth table were achieved. The coherent action of the gate was further demonstrated by generating an atom-photon Bell state $|\Phi^+\rangle$ from a separable input, reaching a fidelity of $(65.90 \pm 1.73)\,\%$ – above the $50\,\%$ threshold required to confirm entanglement generation. The lower Bell-state fidelity compared to the truth-table overlap reflects the additional sensitivity of F_{Bell} to phase errors, which do not contribute to the classical truth-table overlap. A detailed attribution of the infidelity to specific error sources is left to future work. Scans of the overlap against both the V-mode attenuation and the mean photon number agreed with the master-equation simulation, apart from an enhanced drop at $\bar{n} \gtrsim 1$ whose origin remains to be identified. The dominant contributions to the gate infidelity were identified as the asymmetry between the coupled and uncoupled π-mode reflectivities ($\sim 5\%$) and the imperfect atomic state preparation of the coupled atomic state ($\sim 6\%$). The former is set by the current cavity cooperativity and imposes a ceiling on the achievable overlap for the given cavity parameters, whereas the latter can be reduced by refining the individual components of the state-preparation sequence. Smaller contributions that limit the gate overlap include thermal drifts of the detection basis and run-to-run fluctuations of the atom-cavity coupling.

With the truth tables confirming the correct input-output behavior and the generation of a Bell state confirming the coherent action of the gate, the natural next step is to improve the Bell state fidelity through a more systematic optimization of the experimental parameters – in particular, the atomic state preparation fidelity and the photon polarization contrast. Beyond that, inter-node protocols such as quantum state teleportation via a fiber link between the MPQ in Garching and the LMU in Munich come within reach [28]. Another compelling direction is the investigation of nonlinear photon-photon interactions mediated by the single atom, as in the photon-photon gate proposed by Duan and Kimble [22]. A related scheme enabled by the existing infrastructure is the generation of photon-photon entanglement by combining the existing quantum memory capability with the gate demonstrated here: a first photon incident via the long cavity is mapped onto the atomic state, and a second photon is subsequently reflected off the atom-cavity system, producing entanglement between the two temporally separated photons. In contrast to the sequential two-photon reflection scheme of Hacker *et al.* [31], the storage of the first photon provides a heralding signal that confirms its successful arrival before the second photon is sent, allowing the protocol to be conditioned on a verified first photon.

Appendix A

Numerical Simulation of the Atom-Photon Gate

This appendix describes the time-domain master-equation simulations used to determine the optimal Brewster window attenuation (section A.2) and to characterize the gate fidelity as a function of mean photon number (section A.3). All simulations were implemented in Python using the QuTiP framework [48].

A.1 Simulation Model

The simulation models the full open quantum system consisting of a single ^{87}Rb atom coupled to a birefringent Fabry-Pérot cavity supporting two polarization modes (π and V). The atom is described by a four-level Hilbert space spanned by $\{|0\rangle, |1\rangle, |\text{dark}\rangle, |e\rangle\}$, where $|0\rangle$ and $|1\rangle$ are the clock qubit states, $|\text{dark}\rangle$ captures off-resonant decay channels, and $|e\rangle$ is the excited state coupled to $|1\rangle$ via the cavity mode. Each cavity mode is truncated to a Fock space of dimension N_{ph}, yielding a total Hilbert space of dimension $N_{\text{ph}} \times N_{\text{ph}} \times 4$.

A.1.1 Hamiltonian

In the rotating frame, the system Hamiltonian reads

$$\hat{H}_0 = \Delta_{c,\pi}\, \hat{a}_\pi^\dagger \hat{a}_\pi + (\Delta_{c,\pi} + \Delta_{\text{biref}})\, \hat{a}_V^\dagger \hat{a}_V + \Delta_a\, \hat{\sigma}^\dagger \hat{\sigma}, \tag{A.1}$$

with the atom–cavity interaction

$$\hat{H}_{\text{int}} = g\big(\hat{a}_\pi \hat{\sigma}^\dagger + \hat{a}_\pi^\dagger \hat{\sigma} + \hat{a}_V \hat{\sigma}^\dagger + \hat{a}_V^\dagger \hat{\sigma}\big), \tag{A.2}$$

where g is the vacuum coupling rate (identical for both modes), $\Delta_{c,\pi}$ is the cavity–laser detuning for the π mode, $\Delta_{\text{biref}} = 2\pi \times 500\,\text{MHz}$ is the birefringent splitting, and Δ_a is the atom–laser detuning.
The external drive is included via input–output theory. For the CNOT basis states $|\pm\rangle = (|V\rangle \pm |\pi\rangle)/\sqrt{2}$, both cavity modes are driven simultaneously with the appropriate relative phase and amplitude. The time-dependent drive Hamiltonian takes the form

$$\hat{H}^{(\pm)}_{\text{drive}}(t) = \frac{i\,\mu_{fc}\sqrt{\kappa_{OC}}}{\sqrt{2}}\, \varepsilon(t) \Big[\big(\hat{a}_\pi - \hat{a}_\pi^\dagger\big) \pm \sqrt{T_V}\big(\hat{a}_V - \hat{a}_V^\dagger\big) \Big], \tag{A.3}$$

where $\varepsilon(t)$ is the temporal envelope of the input pulse, μ_{fc} is the fiber–cavity mode-matching parameter, κ_{OC} is the outcoupling rate, and T_V is the V-mode power transmission set by the Brewster window array.

A.1.2 Initial States and Imperfect State Preparation

The initial state of the system is a product of the atomic state and the cavity vacuum. For the atomic qubit state $|0\rangle$, the atom is initialized in a pure state $\hat{\rho}_0 = |0\rangle\langle 0|$. To account for imperfect optical pumping and microwave transfer, the nominal $|1\rangle$ initial state is modeled as a density matrix mixture

$$\hat{\rho}_1 = p_1 |1\rangle \langle 1| + (1 - p_1) |0\rangle \langle 0| , \tag{A.4}$$

with $p_1 = 0.88$. The full initial state for each simulation run is then $\hat{\rho}_{\text{init}} = \hat{\rho}_{\text{atom}} \otimes |0_\pi\rangle \langle 0_\pi| \otimes |0_V\rangle \langle 0_V|$.

A.1.3 Dissipation

The open-system dynamics are governed by the Lindblad master equation

$$\dot{\hat{\rho}} = -i[\hat{H}, \hat{\rho}] + \sum_k \mathcal{D}[\hat{L}_k]\hat{\rho}, \tag{A.5}$$

with collapse operators accounting for:

- Photon loss through the outcoupling mirror: $\hat{L}_{1,2} = \sqrt{\kappa_{OC}}\, \hat{a}_{\pi,V}$,
- Internal cavity losses (absorption, HR mirror leakage): $\hat{L}_{3,4} = \sqrt{\kappa_{\text{int}}}\, \hat{a}_{\pi,V}$, where $\kappa_{\text{int}} = \kappa - \kappa_{OC}$,
- Spontaneous emission back to the coupled state: $\hat{L}_5 = \sqrt{p_{e\to 1}\,\gamma}\; |1\rangle \langle e|$,
- Spontaneous emission to off-resonant states: $\hat{L}_6 = \sqrt{p_{e\to\text{dark}}\,\gamma}\; |\text{dark}\rangle \langle e|$,

with branching ratios $p_{e\to 1} = 1/15$ and $p_{e\to\text{dark}} = 14/15$ for the $F' = 1$ excited state of the D_2 line.

A.1.4 Output Fields and Detection Probabilities

The quantum output photon number in each polarization mode is computed via the input–output relation. The mean output field amplitude is

$$\alpha_{\text{out},\alpha}(t) = \mu_{fr}\, \alpha_{\text{in},\alpha}(t) + \mu_{fc}\sqrt{\kappa_{OC}}\, \langle \hat{a}_\alpha(t)\rangle, \tag{A.6}$$

where μ_{fr} is the free-space mode-matching parameter. The quantum output photon number additionally includes the contribution from intracavity fluctuations:

$$n^{(q)}_{\text{out},\alpha}(t) = |\alpha_{\text{out},\alpha}(t)|^2 + |\mu_{fc}|^2 \kappa_{OC} \left(\langle \hat{a}^\dagger_\alpha \hat{a}_\alpha \rangle - |\langle \hat{a}_\alpha \rangle|^2 \right). \tag{A.7}$$

For the CNOT basis, the output fields are transformed via $n^{(q)}_{\text{out},\pm} = (n^{(q)}_{\text{out},V} \pm n^{(q)}_{\text{out},\pi})/2$.
The detection probabilities at the two SNSPD detectors include corrections for dark counts and finite detection efficiency η.

A.1.5 CNOT Truth Table Construction

The CNOT truth table is a 4×4 matrix whose columns correspond to the four input configurations $\{|1,+\rangle, |1,-\rangle, |0,+\rangle, |0,-\rangle\}$ and whose rows correspond to the four output configurations $\{|1,+\rangle, |1,-\rangle, |0,+\rangle, |0,-\rangle\}$, where the first label denotes the atomic state and the second label the photonic polarization. Each entry is the joint probability of observing a given photonic detection outcome *and* the atom being found in a given qubit state after the gate pulse. Concretely, the entry for output $|a', p'\rangle$ given input $|a, p\rangle$ reads

$$\text{CNOT}_{(a',p'),(a,p)} = P_{\text{atom}}(a' \,|\, a, p) \;\cdot\; P_{\text{det}}(p' \,|\, a, p), \tag{A.8}$$

where $P_{\text{atom}}(a' \,|\, a, p)$ is the probability that the atom occupies qubit state $|a'\rangle$ at the end of the pulse (extracted from the final density matrix), and $P_{\text{det}}(p' \,|\, a, p)$ is the photonic detection probability in output port p'. Each column is then normalized to unit probability.
This construction ensures that spontaneous scattering events—which can flip the atomic qubit state during the gate pulse—are reflected as errors in the truth table, rather than being accounted for as a separate correction factor.
The average probability for the atom to remain in its initial qubit state is

$$p_{\text{atom}} = \frac{1}{4} \sum_{a\in\{0,1\},\, p\in\{+,-\}} P_{\text{atom}}(a \,|\, a, p), \tag{A.9}$$

which provides a direct measure of the atomic state preservation across all input configurations.

A.1.6 Simulation Parameters

The parameters used in the simulations are summarized in Table A.1. These correspond to the experimentally characterized values presented in section 5.1.

Parameter	Symbol	Value
Atom–cavity coupling	g	$2\pi \times 25.9\,\mathrm{MHz}$
Total cavity decay rate	κ	$2\pi \times 58\,\mathrm{MHz}$
Outcoupling ratio	κ_{OC}/κ	0.85
Atomic decay rate	γ	$2\pi \times 6.065\,\mathrm{MHz}$
Birefringent splitting	Δ_{biref}	$2\pi \times 505\,\mathrm{MHz}$
Fiber–cavity mode-matching	μ_{fc}	0.873
Free-space mode-matching	μ_{fr}	0.978
State preparation fidelity	p_1	0.88
Detection efficiency	η	0.742
Dark count rate	r_{dc}	5×10^{-5}
Off-resonant detuning to $F' = 3$	$\Delta_{F'3}$	$2\pi \times 423.6\,\mathrm{MHz}$
Free-space scattering rate coefficient	κ_{fs}	$2\pi \times 58\,\mathrm{MHz}$

Table A.1: Parameters used in the master-equation simulations. The detection efficiency is the product of fiber coupling (0.9), SNSPD efficiency (0.85), and optical path transmission (0.97).

A.2 Brewster Window Transmission Scan

To determine the optimal number of Brewster windows for the V-mode attenuation (cf. section 4.1), the gate fidelity was computed as a function of the V-mode transmission $T_V \in [0, 1]$.
For each value of T_V, the simulation was run in the CNOT basis: weak coherent pulses with $|+\rangle$ and $|-\rangle$ input polarizations were reflected from the atom-cavity system for both atomic qubit states $|0\rangle$ and $|1\rangle$. In the simulation, the V-mode transmission enters as a prefactor $\sqrt{T_V}$ on the V-mode drive amplitude, effectively modeling the attenuation introduced by the Brewster window array prior to the cavity. The Fock space truncation was set to $N_{\mathrm{ph}} = 3$ for this scan.
From the four resulting output fields, the 4×4 CNOT truth table was constructed as described in section A.1.5, including the atomic state probability weighting. The columns of the truth table were then normalized to unit probability. The overlap was evaluated as the column-averaged overlap between the simulated and ideal CNOT truth tables:

$$\mathrm{Overlap} = \frac{1}{4} \sum_{j=1}^{4} \left| \langle U_j | V_j \rangle \right|^2 , \tag{A.10}$$

where $|U_j\rangle$ and $|V_j\rangle$ are the j-th columns of the ideal and simulated truth tables, respectively.
The scan shown in Figure A.1 reveals a clear optimum at $T_{\mathrm{V}} \approx 0.24$, corresponding to a V-mode power attenuation of approximately 76 %. Given the measured single-window V-mode transmission of $(77.40 \pm 0.13)\,\%$ (section 5.1.2), the naïve estimate 0.7738^n predicts an optimum between 5 and 6 Brewster windows ($0.7740^5 \approx 0.277$ and $0.7740^6 \approx 0.215$). However, the directly measured multi-window transmissions of $(26.80 \pm 0.06)\,\%$ for 6 windows and $(22.47 \pm 0.04)\,\%$ for 7 windows bracket T^*_{V}, indicating that 6 to 7 Brewster windows realise the optimal V-mode attenuation in practice.

A.3 Mean Photon Number Scan

While the atom-photon gate is designed to operate in the single-photon regime, the use of weak coherent pulses introduces a non-zero probability of multi-photon events that degrade the gate. This section characterizes how the gate overlap depends on the mean photon number $\bar{n}$ of the input pulses, and compares the simulation to the experimental measurement.

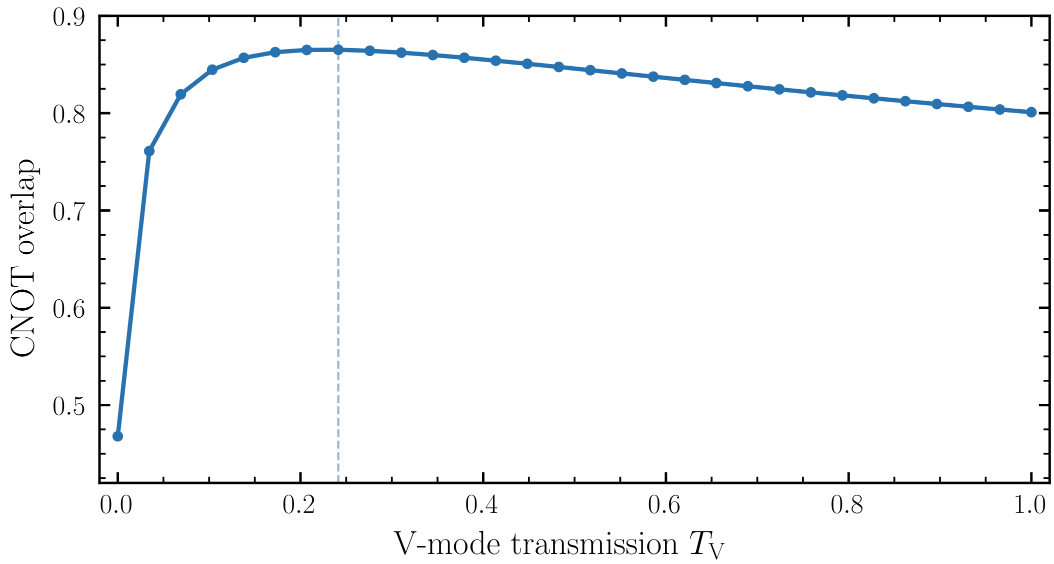


Figure A.1: Simulated CNOT gate overlap as a function of the V-mode cavity transmission T_{V}, obtained from a master-equation simulation of the atom–photon gate protocol. The dashed line marks the optimal transmission $T_{\mathrm{V}}^* \approx 0.24$.

The mean photon number was scanned over the range $\bar{n} \in [10^{-5}, 10]$ on a logarithmic scale. For each value of $\bar{n}$, the coherent pulse amplitude was computed from the pulse normalization as

$$\alpha = \sqrt{\frac{\bar{n}}{\int_0^T |\varepsilon_{\mathrm{ref}}(t)|^2 \, dt}}, \tag{A.11}$$

where $\varepsilon_{\mathrm{ref}}(t)$ is the unit-amplitude pulse envelope. The CNOT truth table was then constructed as described in section A.1.5, including the atomic state probability weighting, and the overlap with the ideal CNOT truth table was evaluated using (A.10). The V-mode transmission was fixed at $T_V = 0.2247$ and the Fock space truncation was set to $N_{\mathrm{ph}} = 3$ for this scan.
The result is shown in Figure A.2. The simulation captures three distinct regimes:
At very low photon numbers ($\bar{n} \lesssim 10^{-3}$), dark counts dominate the detected signal and the overlap approaches 0.5, corresponding to random guessing. As $\bar{n}$ increases, the signal-to-noise ratio improves and the overlap rises towards a plateau at $\bar{n} \approx 10^{-1}$, where the gate operates near its single-photon limit with an overlap of ≈ 0.865. For $\bar{n} \gtrsim 1$, multi-photon events begin to degrade the overlap, as the master-equation simulation captures the nonlinear response of the atom-cavity system to multi-photon input states.

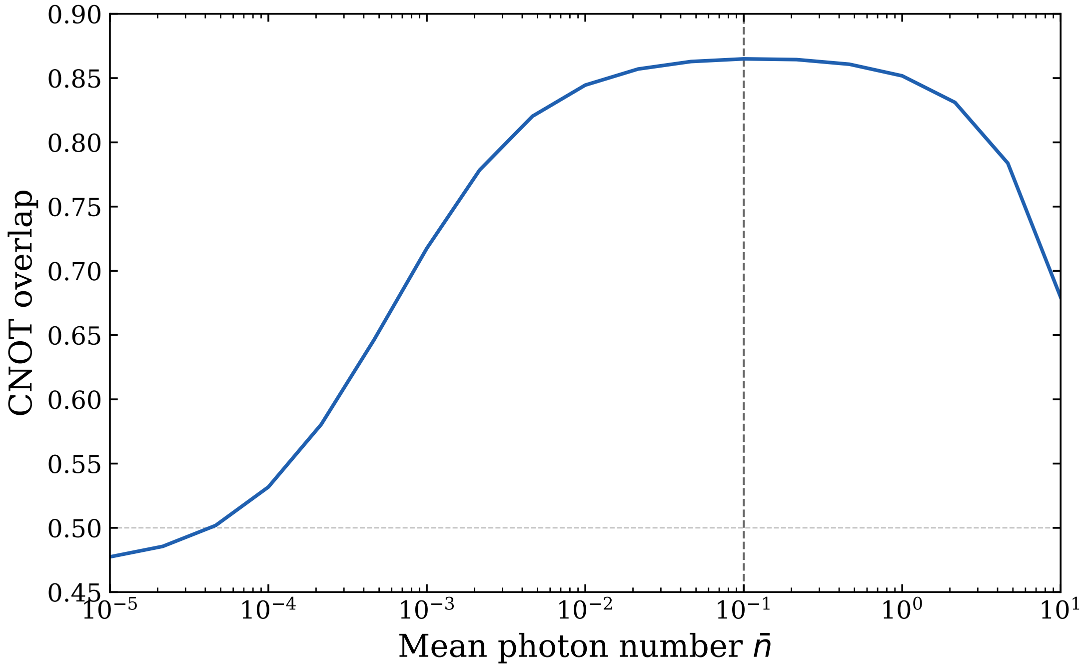


Figure A.2: Simulated CNOT gate overlap (blue line) as a function of the mean input photon number $\bar{n}$. The V-mode transmission is fixed at $T_V = 0.2247$ and the Fock space is truncated at $N_{\mathrm{ph}} = 3$. The simulation reproduces the experimental overlap across six orders of magnitude in $\bar{n}$, with a maximum overlap of ≈ 0.865 at $\bar{n} \approx 0.1$.

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